A fixed-target approach to the sharp iid Berry–Esseen inequality*

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Abstract

We study the conjectured sharp iid Berry–Esseen constant by maximizing excess over its fixed target. Attainment and contact give a common moment-defect budget; a specified Fourier certificate supplies coarse entry. For $n \geq 700$, direct geometry and two contact rays acquire one support strip. A single-count fourth-moment bound handles small noise, and a five-count identity handles the rest. Both comparisons preserve the actual marginal, threshold atoms and complete tails. The resulting global implication remains conditional on the stated entry predicates and a genuine finite base through 699, which is not proved here.

1 The sharp constant and the proof route

For centered, variance-one laws P with $\rho(P) = \mathbb{E}|X|^3 < \infty$, let S_n be a sum of independent copies and set

$$\Delta_n(P) = \sup_{H \in \mathbb{R}} |\mathbb{P}_P(S_n \leq H) - \Phi(H/\sqrt{n})|, \qquad C_n = \sup_P \frac{\sqrt{n}\Delta_n(P)}{\rho(P)}.$$

The iid sharp-constant conjecture is

$$\sup_{n \geq 1} C_n = C_{\mathrm{E}}, \qquad C_{\mathrm{E}} = \frac{3 + \sqrt{10}}{6\sqrt{2\pi}}. \tag{1}$$

Here Φ, ϕ denote the standard normal distribution function and density; H is always a raw sum threshold. Finite decimals mean exact rationals. Affine standardization gives the usual arbitrary-mean, positive-variance form.

The classical Berry–Esseen theorem gives the order $n^{-1/2}$ [1, 2]. Esseen's fixed Bernoulli example gives the sharp lower approximation [3]. Namely, for $p_* = (4 - \sqrt{10})/2$, $q_* = 1 - p_*$ and $X = (B - p_*)/\sqrt{p_* q_*}$,

$$\lim_{n \to \infty} \frac{\sqrt{n}\Delta_n(P)}{\rho(P)} = \frac{3 + q_* - p_*}{6\sqrt{2\pi}(p_*^2 + q_*^2)} = C_{\mathrm{E}}. \tag{2}$$

The last equality uses $p_*^2 + q_*^2 = 10 - 3\sqrt{10}$. The universal upper conjecture is associated with Zolotarev [6]. We use Shevtsova's structural moment bound [5] and Schulz's strict finite-binomial theorem [4], both stated where applied.

*This paper was written with substantial assistance from AI systems; see the end of Section 7.

Fix $c = C_{\mathrm{E}}$ and maximize the excess

$$a(P, H) = \sqrt{n}\{\mathbb{P}_P(S_n \le H) - \Phi(H/\sqrt{n})\} - c\rho(P). \tag{3}$$

The fixed moment penalty gives attainment and compact-support contact. Integrating contact against the zero-bias density gives one budget for excess, moment defect and leakage. Complete coarse Fourier entry supplies $nD \le 32.20665378$. At $n \ge 700$, direct geometry and two contact rays then acquire one probability/support strip. Two comparisons, split at noise $\lambda = 1/100$, exclude its nonnegative excess.

All steps retain the same actual marginal. Reflection chooses its positive closed-event orientation before signed acquisition. The local comparison itself requires neither contact nor a predecessor; these are used only to acquire its hypotheses. The global implication in Section 7 still needs a genuine base through 699: a base only through 41 leaves $42, \dots, 699$ unresolved.

2 Fixed-target extremality and conservation

Fix $n \ge 2$ and $c > 0$. For a standardized marginal with finite third absolute moment, write

$$z = H/\sqrt{n}, \quad F = \mathbb{P}(S_n \le H), \quad a = \sqrt{n}\{F - \Phi(z)\} - c\rho,$$
$$\varphi = \phi(z), \quad \beta = \mathbb{E}[X|X|], \quad \gamma = \mathbb{E}X^3.$$

The threshold H is raw, and a is not divided by ρ.

Proposition 2.1 (Attainment and contact)**.** *The supremum of a over standardized marginals and thresholds is attained. Every attaining pair has positive closed-CDF discrepancy and compact support. On the whole line,*

$$\begin{aligned} I(y) = n^{3/2}\{h(y) - F\} + ny\varphi + \tfrac{1}{2}H\varphi(y^2 - 1) \\ + \tfrac{3}{2}c\rho(y^2 - 1) - c\{|y|^3 - \rho - 3y\beta\} \le 0, \qquad h(y) = \mathbb{P}(S_{n-1} \le H - y), \end{aligned} \tag{4}$$

with equality on the entire closed support. No sign assumption on a or H is needed.

Proof. The supremum A satisfies $-c < A \le \sqrt{n} - c$: the lower inequality uses symmetric Bernoulli at $H = n$. A maximizing sequence therefore has bounded ρ_j and a positive discrepancy bounded away from zero. Chebyshev bounds the negative threshold tail by n/H_j^2, and the positive discrepancy is bounded by the normal tail. Hence H_j is bounded. Pass to $P_j \Rightarrow P$, $H_j \to H$, $\rho_j \to b$. A common third-moment ceiling M gives

$$\int_{|x|>K} x^2 dP_j \le M/K, \qquad \int_{|x|>K} |x| dP_j \le M/K^2.$$

Truncation preserves mean and variance, while lower semicontinuity gives $1 \le \rho(P) \le b$. The sum laws converge weakly; apply the closed-set inequality at $H + t$ and let $t \downarrow 0$ to obtain $\limsup F_j(H_j) \le F_P(H)$, including atoms. Thus $a(P, H) \ge A + c\{b - \rho(P)\} \ge A$. Maximality gives attainment and $\rho(P) = b$; third-moment convergence has not been assumed. The discrepancy is positive because $A + c\rho \ge A + c > 0$.

Contaminate by $(1 - \epsilon)P + \epsilon\delta_y$, then center and scale, evaluating at the standardized image of the same raw event. The mean and variance are ϵy and $1 + \epsilon(y^2 - 1) - \epsilon^2 y^2$. The

derivatives at zero of the raw event probability, centered third absolute moment and normal argument are, respectively,

$$n(h(y) - F), \qquad |y|^3 - \rho - 3y\beta, \qquad -\sqrt{n}y - H(y^2 - 1)/(2\sqrt{n}).$$

The event probability is a polynomial even at collisions; dominated moment differentiation uses $3|y|(|x| + |y|)^2$. Differentiating the excess gives $I(y) \le 0$ and $\int I\, dP = 0$. The closed-event function h, hence I, is upper semicontinuous. Strict negativity at a support point would persist on a positive-mass neighborhood, contradicting that integral. Finally $I(y) \le -c|y|^3 + O(y^2 + |y| + 1)$, so contact excludes both distant rays. □

Reflection and first failure.

For every law, including an atom at H,

$$\mathbb{P}(-S_n \le -H) - \Phi(-z) = \Phi(z) - \mathbb{P}(S_n < H) \ge \Phi(z) - \mathbb{P}(S_n \le H).$$

Thus C_n is also the supremum of positive closed scores r. Since $a = \rho(r - c)$, $C_n < c$ implies $A \le C_n - c < 0$, and $C_n > c$ implies $A > 0$. If $C_n = c$, choose $r_j \uparrow c$ with $r_j \ge c/2$; then $\rho_j \le 2\sqrt{n}/c$ gives $a_j \to 0$, so $A = 0$. Consequently $C_n \ge c$ supplies an attained $a \ge 0$ witness. If a strict bound fails above a genuine weak base $C_j \le c$ for $j < N$, the least failing index has this witness and true predecessors $C_j \le c$ for all $j < n$.

2.1 An identity that retains contact residuals

Write $\Delta = Q - h$, where contact is equivalent to $\Delta \ge 0$ and $\Delta = 0$ on the closed support, and

$$\begin{aligned} Q(y) = F + \frac{H\varphi + c\rho}{2n^{3/2}} - \frac{n\varphi + 3c\beta}{n^{3/2}}y \\ - \frac{H\varphi + 3c\rho}{2n^{3/2}}y^2 + \frac{c|y|^3}{n^{3/2}}. \end{aligned} \tag{5}$$

For now allow any compact standardized law, without contact. Put $\nu = \mathcal{L}(H - S_{n-1})$ and $\tau(t) = \mathbb{E}[X\mathbf{1}_{X \ge t}]$. Then $\tau \ge 0$, $\int \tau = 1$. On each bounded gap (u, v) of the closed support K, its interior value is a constant $\tau_{u,v}$. Define

$$J = \sum \tau_{u,v}(v - u)^2, \qquad J_1 = \sum \tau_{u,v} \int_u^v (t - u)t\, dt,$$
$$J_2 = \sum \tau_{u,v} \int_u^v (t - u)t|t|\, dt, \qquad L = \sum \tau_{u,v} \int_{[u,v)} (t - u)\, d\nu(t).$$

The residuals are

$$\xi = \mathbb{E}[X\Delta(X)], \qquad B = \int_K \tau\Delta + \sum \tau_{u,v}(v - u)\Delta(v). \tag{6}$$

The first integral is over Lebesgue measure on K and cannot be omitted. Compactness makes these quantities finite; the residual sum converges absolutely since Δ is bounded and $\sum \tau_{u,v}(v - u) \le 1$. Let $M_4 = \mathbb{E}X^4$, $M_5 = \mathbb{E}|X|^5$ and set

$$K_q = M_5/4 - \rho M_4/2 - 3\beta\gamma/2 + \rho/2, \qquad K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2,$$
$$\Omega = J_1 + (M_4 - 3)/6, \qquad q_z(x) = (x - z)|x - z|,$$
$$\varepsilon = \mathbb{E}q_z(S_n/\sqrt{n}) - \mathbb{E}q_z(Z_0) - 2\gamma\varphi/(3\sqrt{n}), \qquad E_n = -n^{3/2}\varepsilon/2,$$

where Z_0 is standard normal. Define

$$\mathcal{R}_c = c(3\beta J/2 + 3\rho J_1 - 3J_2 - K_q) + H(cK_1 + \varphi\Omega) - H^2(c\rho + \gamma\varphi/2) + E_n. \tag{7}$$

Theorem 2.2 (Conservation)**.** *For every compact standardized law,*

$$(n + H^2)a + nc\rho - \frac{n\varphi}{6}(\gamma + 3J) + n^{3/2}L = \mathcal{R}_c + n^{3/2}(B - H\xi). \tag{8}$$

At an attaining pair the residuals vanish, so

$$(n + H^2)a + nc\rho - \frac{n\varphi}{6}(\gamma + 3J) + n^{3/2}L = \mathcal{R}_c. \tag{9}$$

Proof. Evaluate $\mathcal{A} = \int \tau\Delta$ twice. Inside a gap, $h(t) = h(v) + \nu([t, v))$, whence Fubini gives

$$\int_u^v \Delta = (v - u)\Delta(v) - \int_u^v (t - u)Q'(t)dt - \int_{[u,v)} (t - u)d\nu(t).$$

The atom at v belongs to $h(v)$, not the last integral; the atom at u has zero weight. Summing with the support integral yields

$$\mathcal{A} = \frac{\varphi J}{2\sqrt{n}} + \frac{3c\beta J/2 + (H\varphi + 3c\rho)J_1 - 3cJ_2}{n^{3/2}} - L + B. \tag{10}$$

For the other evaluation use signwise Fubini, $\int \tau g' = \mathbb{E}[Xg(X)]$. In particular $\int t\tau = \gamma/2$, $\int t^2\tau = M_4/3$ and $\int |t|^3\tau = M_5/4$. Condition on S_{n-1} and take $g(x) = \min(S_{n-1} + x, H)$; exchangeability gives

$$\int \tau h = \mathbb{E}[X_n \min(S_n, H)] = \frac{1}{n}\mathbb{E}[S_n \min(S_n, H)] = T := \mathbb{E}[(Z^2 - zZ)\mathbf{1}_{Z\le z}], \qquad Z = S_n/\sqrt{n}.$$

The endpoint integrand is zero, so atoms cause no extra term. Consequently

$$\mathcal{A} = F - \frac{\gamma\varphi}{2\sqrt{n}} + \frac{H\varphi(3 - M_4)/6 + cK_q}{n^{3/2}} - T. \tag{11}$$

Put $M = \mathbb{E}[Z\mathbf{1}_{Z\le z}]$. Integrating $XQ(X)$ against P gives

$$M + \varphi = -\gamma H\varphi/(2n) + cK_1/n - \sqrt{n}\xi. \tag{12}$$

The pointwise identity $q_z(x) = (x - z)^2 - 2(x - z)^2\mathbf{1}_{x\le z}$, also applied to Z_0, gives

$$T - \Phi(z) = -\varepsilon/2 - \gamma\varphi/(3\sqrt{n}) + z(M + \varphi) - z^2(F - \Phi(z)).$$

Insert this and (12) into (11), equate with (10), and use $\sqrt{n}(F - \Phi(z)) = c\rho + a$. Collecting terms proves (8), including the coefficient $n + H^2$. Closed-support contact makes both residuals zero. □

2.2 The resource and the order of its use

The gap bounds proved in Lemma 3.1 give

$$D = (3 + \sqrt{10})\rho - \gamma - 3J \ge 0. \tag{13}$$

At $c = C_{\mathrm{E}} = (3 + \sqrt{10})\phi(0)/6$, let $w = H^2/(2n)$, $e = e^{-w}$. Conservation becomes

$$\mathcal{R}_c = (n + H^2)a + nc\rho(1 - e) + \frac{n\phi(0)e}{6}D + n^{3/2}L. \tag{14}$$

Every term is nonnegative when $a \ge 0$.

Corollary 2.3 (Raw ceiling and phase price)**.** *At an attaining pair with $a \geq 0$, suppose $n \geq N \geq 2$, $\rho \geq l > 0$, $0 < c_- \leq c$, and $\mathcal{R}_c \leq T < Nc_-l$. Negative T excludes the pair. Otherwise*

$$\gamma + 3J > 0, \qquad e \geq 1 - T/(Nc_-l) > 0,$$
$$\mathcal{U} := (n + H^2)a + n\phi(0)D/6 + n^{3/2}L \leq \mathcal{R}_c - \tfrac{1}{2}(c_-l - T/N)H^2 \leq \mathcal{R}_c.$$

Proof. Set $M_c = c\rho - \phi(0)D/6 = \phi(0)(\gamma + 3J)/6$. The identity gives $eM_c = (1 + H^2/n)a + c\rho + \sqrt{n}L - \mathcal{R}_c/n \geq c\rho - T/n > 0$. Since $D \geq 0$, $M_c \leq c\rho$, proving the sign and density claims. Also $\mathcal{R}_c - \mathcal{U} = n(1-e)M_c$, and

$$n(1-e)M_c = n(e^w - 1)eM_c \geq \tfrac{1}{2}H^2(c\rho - T/n) \geq \tfrac{1}{2}H^2(c_-l - T/N).$$

Thus a raw ceiling licenses the density floor before the resource is improved. The improved resource is not a new raw ceiling. □

3 The sign split and the physical gap defect

For a standardized law put $p = \mathbb{P}(X > 0)$, $q = 1 - p$, $v = pq$ and $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$. Both sign classes have positive mass; any zero atom belongs to the nonpositive class. With $B_0 = \mathbf{1}_{X>0}$, set

$$V = \mathbb{E}(X \mid B_0), \quad W = X - V, \quad \theta = \mathbb{E}W^2 = 1 - m^2/v,$$
$$\ell = m/v = \sqrt{(1-\theta)/v}, \quad U = W/\ell.$$

The conditional means are $-m/q$ and m/p, so exactly

$$X = \ell(B_0 + U - p), \qquad \mathbb{E}(U \mid B_0) = 0, \quad s = \mathbb{E}U^2 = \theta/\ell^2, \quad \lambda = ns, \quad v + s = \ell^{-2}. \tag{15}$$

If $s = 0$ the law is Bernoulli. Otherwise let U_i be the conditional noises and $w_i = \mathbb{E}U_i^2$, $\mu_i = \mathbb{E}U_i^3$. Expanding each actual sign class, with conditional centering, gives

$$\begin{aligned}
\rho/\ell^3 &= v(1-2v) + 3v(w_0 + w_1) - q\mu_0 + p\mu_1,\\
\gamma/\ell^3 &= v(q-p) + 3v(w_1 - w_0) + q\mu_0 + p\mu_1,\\
\beta/\ell^2 &= v(q-p) + pw_1 - qw_0.
\end{aligned} \tag{16}$$

These are identities for the same marginal, not independently chosen moment relaxations.

Lemma 3.1 (Moment and gap bounds)**.** *For $\sigma = \sqrt{10}$ and the actual J of Section 2,*

$$\gamma^2 + 4m^2 \leq \rho^2, \quad J \leq \rho + 2m, \quad (2\rho)^{-1} \leq m \leq \tfrac{1}{2}, \tag{17}$$

$$|\beta| < 1, \qquad \rho \geq \frac{1+\beta^2}{\sqrt{1-\beta^2}}, \qquad D := (3+\sigma)\rho - \gamma - 3J \geq 0. \tag{18}$$

If zero is in the closed support, $D \geq (\sigma - 1)\rho$. Otherwise its central gap $(-A, B)$ satisfies $A, B > 0$, $AB \leq 1$ and $J \leq \rho + 2mAB$.

Proof. Let $a_\pm = \mathbb{E}X_\pm^2 = (1 \pm \beta)/2$ and $b_\pm = \mathbb{E}X_\pm^3 = (\rho \pm \gamma)/2$. Signwise Cauchy gives $a_\pm^2 \leq mb_\pm$ and $m^2 \leq pa_+$, $m^2 \leq qa_-$. Hence $m^2 \leq a_+a_-$ and $b_+b_- \geq m^2$, proving the circle

inequality. Also $m\rho \geq a_+^2 + a_-^2 = (1+\beta^2)/2$ and $2m \leq \sqrt{1-\beta^2}$ give the signed moment bound. The bounds for m follow from $\mathbb{E}|X| \leq 1$ and $1 \leq \sqrt{\mathbb{E}|X|\mathbb{E}|X|^3} = \sqrt{2m\rho}$.

Signwise Tonelli gives

$$\int \tau = 1, \qquad 2\int |t|\tau(t)dt = \rho. \tag{19}$$

Each gap on one side of zero obeys $(b-a)^2 \leq 2\int_a^b |t|dt$. A central gap has weight m and excess $2mAB$ over this integral; $0 \leq \mathbb{E}[(X+A)(X-B)] = 1 - AB$. Summing proves the bounds for J, also for countably many gaps. Cauchy in $\mathbb{R}^2$ gives $\gamma + 6m \leq \sqrt{10}\rho$, and therefore $D = 3(\rho + 2m - J) + (\sigma\rho - \gamma - 6m) \geq 0$. Without a central gap, $J \leq \rho$ gives $D \geq (\sigma - 1)\rho$. □

For the central gap define

$$T_+ = \mathbb{E}X_+^3 - mB^2, \quad T_- = \mathbb{E}X_-^3 - mA^2, \quad R_+ = \mathbb{E}X_+^2 - mB, \quad R_- = \mathbb{E}X_-^2 - mA.$$

All four quantities are nonnegative. If $J_o = J - m(A+B)^2$, Tonelli and the positive outer-gap inequality give

$$J_o^+ \leq 2\int_B^\infty (t-B)\tau(t)dt = \mathbb{E}[X_+(X_+ - B)^2] = T_+ - 2BR_+;$$

the negative counterpart is $J_o^- \leq T_- - 2AR_-$. Expanding D yields

$$\begin{aligned} D = m\{\sqrt{\sigma+1}A - \sqrt{\sigma-1}B\}^2 + (\sigma-1)T_+ + (\sigma+1)T_- \\ + 3(T_+ + T_- - J_o). \end{aligned} \tag{20}$$

Every term is nonnegative. Thus $T_+ + T_- \leq D/2$ and, if $D < 2$, $m(A^2+B^2) = \rho - T_+ - T_- \geq 1 - D/2$ gives $A + B \geq \sqrt{2-D}$. The stronger outer-gap bounds also give the useful joint inequality

$$D \geq (\sigma-1)\mathbb{E}X_+^3 + (\sigma+1)\mathbb{E}X_-^3 - 6mAB + 6BR_+ + 6AR_-.$$

This retains the actual gap and conditional noises before any costs are weakened. It is the common starting point of the direct acquisition.

4 Coarse entry from contact

Fix an attained pair with $a \geq 0$, $n \geq N = 42$, and true predecessor $C_{n-1} \leq c = C_{\mathrm{E}}$. All bounds below concern this one marginal and raw threshold. They pass through two quadratic and four signed updates; their finite coverage conditions remain explicit.

4.1 Support before higher moments

Theorem 4.1 (Initial entry)**.** *Assume Shevtsova's external structural inequality [5]*

$$\sqrt{n}\Delta_n(P) \leq \frac{3031}{10000}(\rho + 323/500). \tag{21}$$

Then $\rho < 7361/4000$, $|\beta| < 321/500$ *and* $\operatorname{supp} P \subset (-96/25, 96/25)$, *uniformly in* H.

Proof. Since $c\rho \le \sqrt{n}\Delta_n(P)$ and $c > 819/2000$, division gives

$$\rho < \frac{(3031/10000)(323/500)}{819/2000 - 3031/10000} = \frac{7361}{4000}.$$

Signwise Cauchy gives $\rho \ge (1+\beta^2)/\sqrt{1-\beta^2}$, increasing in $|\beta|$. At $r = 1841/1000$, $b = 321/500$,

$$(1+b^2)^2 - r^2(1-b^2) = \frac{93288849}{50000000000} > 0.$$

Put $v = n-1$, $s = \sqrt{n/v}$, $F_v(x) = \Phi(x/\sqrt{v})$ and $e_n = n(s-1) - 1/2$. Contact and the predecessor imply

$$0 \le H_n(H,t) + c\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho e_n\} - na \quad (t \in \operatorname{supp} P), \tag{22}$$

where

$$H_n/n^{3/2} = F_v(H-t) - F_{v+1}(H) + tF'_{v+1}(H) - (t^2-1)F''_{v+1}(H)/2.$$

The extra $-na$ is nonpositive; no unknown-score comparison is used. Rationalization gives $ne_n = s^2(2s+1)/(2(s+1)^2) < 793881/2073680 < 48/125$ and $s < S := 81/80$, hence $e_n < \eta := 8/875$; the rationalized expression increases for $s \ge 1$.

The heat identity $\partial_v F_v = F''_v/2$, spatial Taylor expansion to order four and variance expansion to order two give, for $u \ge 1$,

$$H_n(H, \pm u)/s^3 = \pm A(1-z_v^2)\phi(z_v) + B\phi'''(z_v) + \mathcal{E}, \tag{23}$$

where $z_v = H/\sqrt{v}$, $A = (u^3-3u)/6$, $H_4 = u^4 - 6u^2 + 3$, $B = H_4/(24\sqrt{v})$ and

$$|\mathcal{E}| < E(u) := u^5/4100 + 3u/820 + 17(3u^2-2)/78720.$$

The four remainders after division by s^3 are bounded by

$$\frac{u^5\|\phi^{(4)}\|_\infty}{120v}, \quad \frac{\|\phi^{(5)}\|_\infty}{48v^{3/2}}, \quad \frac{u\|\phi^{(4)}\|_\infty}{8v}, \quad \frac{|u^2-1|\|\phi^{(5)}\|_\infty}{16v^{3/2}}.$$

Gaussian inversion bounds these derivative norms by $6/5$ and $51/20$; use $v \ge 41$, $\sqrt{v} > 6$. No higher moment of P enters. For $A, B > 0$, $B \le A/4$, the leading expression is at most $\phi(0)(A + 9B^2/(4A))$ for both signs: for the positive sign on $|z_v| \le 1$, complete the square in $A - Az_v^2 + 3B|z_v|$; outside it the first term is nonpositive. For the negative sign use $2e^{-3/2}A < A/2$ and $\|\phi'''\|_\infty < 2\phi(0)$. The latter follows from $3x - x^3 \le 2$ on $[0, \sqrt{3}]$ and $x^3 e^{-x^2/2} \le 3\sqrt{3}e^{-3/2} < 2$ beyond it.

Freeze the preceding r, b, and set

$$Q(u) = -u^3 + 3ru^2/2 + 3bu + \eta r,$$

$$P(u) = \tfrac{2}{5}S^3\{A^2 + H_4^2/(256\cdot 41)\} + S^3E(u)A + (819/2000)Q(u)A.$$

At $R = 96/25$ we have $A > 0$, $0 < H_4 < 36A$, $Q(R) < 0$ and

$$-P(R) = \frac{2214258283536204102723}{10250000000000000000000} > \frac{1}{5}.$$

Contact is therefore impossible at either boundary. For the whole rays,

$$|\partial_t H_n(H,t)| < \tfrac{2}{5}(t^2/2 + 1/2 + e_n).$$

Indeed, with $y = (H-t)/\sqrt{n}$, $z = H/\sqrt{n}$, the derivative is $n\phi(z) + Ht\phi(z) - ns\phi(sy)$. The spatial second-order error costs $\phi(0)t^2/2$, while $\partial_s[s\phi(sy)] = -\phi''(sy)$ costs $\phi(0)n(s-1)$. Put $L(u) = \frac{2}{5}(u^2/2 + 1/2 + \eta) + (819/2000)Q'(u)$. At R, all Q, Q', Q'', L, L', L'' are negative. As $Q''' = -6$ and $L'' = 2/5 - 6(819/2000) < 0$, the signs persist outwards, so both actual radial upper expressions decrease from negative boundary values. □

The same heat/ray argument licenses a later radius in $\{2.50, 2.51, \dots, 3.84\}$: freeze the cell bounds r, b, then check $A > 0$, $0 < H_4 < 36A$, $Q(R), P(R) < 0$ and the six ray signs before using that radius in any higher-moment bound.

4.2 The common Fourier input

After acquiring support, put $M_j = \mathbb{E}|X|^j$. The measure $d\mu_0 = (x-1)^2(x+1/2)dP$, $x = |X|$, has mass $\rho - 1$, and

$$M_4 - 1 = \int \frac{(x+1)^2}{x+1/2} d\mu_0, \qquad M_5 - 1 = \int \frac{x^3 + 2x^2 + 3x + 3/2}{x+1/2} d\mu_0.$$

The ratios increase on $x \geq 0$; write k_4, k_5 for their values at R. Consequently

$$M_4 \leq \min\{R\rho, 1 + k_4(\rho - 1)\}, \quad M_5 \leq \min\{RM_4, 1 + k_5(\rho - 1)\}, \quad M_6 \leq RM_5. \tag{24}$$

Also $\gamma^2 \leq M_4 - 1$ by $\mathbb{E}(X^2 - 1 - \gamma X)^2 \geq 0$. For two iid copies, $\mathcal{E} = \{(X - X')^2 \geq 9/25\}$ satisfies

$$\mathbb{P}(\mathcal{E}) \geq \alpha := \frac{(2 - 9/25)^2}{2M + 6}, \quad |X - X'| \geq 3/5 \text{ on } \mathcal{E}, \quad \mathbb{E}[(X - X')^2 \mid \mathcal{E}] \leq 2/\alpha, \tag{25}$$

where $M \geq M_4$. Cauchy proves this from $\mathbb{E}[(X - X')^2; \mathcal{E}] \geq 2 - 9/25$ and $\mathbb{E}(X - X')^4 = 2M_4 + 6$; the last bound uses $\mathbb{E}(X - X')^2 = 2$.

Appendix A proves the complete compensated Fourier bound, including atoms and the comparator tail. Its input is an acquired radius R, ceilings $M \leq 11.1$, $g \leq 1.85$ for $M_4, |\gamma|$, a sixth-moment ceiling, and $|(3 - M_4)/24| \leq A_4$, $|(10\gamma - \mathbb{E}X^5)/120| \leq Z_5$. Use the complete LOW triple L_0, L_H, L_2 of Proposition A.2, with the sharper fifth-order coefficient proved there. An actual event of mass at least α, separation at least d_E and conditional second moment at most M_E supplies the six-term HIGH bound K_H of Proposition A.3, under all its damping and all-index gates. Insert each event option into all six terms before minimization. The output is

$$|E_n| \leq K_F + s_F|H|, \qquad K_F = \tfrac{16}{25}(L_0 + K_H + L_2/\sqrt{N}), \quad s_F = \tfrac{16}{25}L_H. \tag{26}$$

It is uniform in actual $n \geq N$, even when the law and threshold vary.

4.3 Coarse coefficients and two passes

The constant coefficient of c in (7) is

$$A^{\text{law}} = 3\beta(J + \gamma)/2 + 3(\rho J_1 - J_2) + \rho M_4/2 - M_5/4 - \rho/2.$$

Without a central gap one still has

$$-M_4/6 \leq J_1 \leq M_4/2, \qquad -3J_2 - M_5/4 \leq 0, \qquad J \leq \rho + 1. \tag{27}$$

For a gap (u, v) of length h, the first pair follows from $\int(t-u)t + \frac{1}{2}\int t^2 = hv^2/2$ and $\frac{3}{2}\int t^2 - \int(t-u)t = h(2u+v)^2/6$; sum against its weight and use $\int t^2\tau = M_4/3$. On negative gaps reflect and use $\frac{1}{3}\int_u^v x^3dx - \int_u^v(v-x)x^2dx = u^3(v-u)/3 \geq 0$ for the second bound; positive pieces improve it. The last uses $J \leq \rho + 2m$. For $1 \leq r \leq \rho$, $R \leq 6$, choose

$$\Gamma(R, r) \geq \max_{r \leq x \leq R} \left\{-x^2/4 + (R+r)x/3 - Rr/2 + r^3(2R-r)/(12x^2)\right\}_+ . \tag{28}$$

Then

$$3(\rho J_1 - J_2) \leq 3\Gamma(R, r)(\rho - \gamma)/2 + 3\rho^2 J/8. \tag{29}$$

On a negative gap discard $|t| < \rho$, whose charge is nonpositive. Extend the rest to $\int_r^v (R - x)x(x-r)dx$; dividing by v^2 gives (28). Tonelli charges its bound against $\mathbb{E}X_-^3 = (\rho - \gamma)/2$. On positive pieces use $t(\rho - t) \leq \rho^2/4$ and $\sum \tau \int(t-u) = J/2$. To enclose Γ, split $[r, R]$ into 64 closed intervals, maximize the quadratic at its clipped vertex $2(R+r)/3$ and the positive reciprocal term at the left endpoint. Alternatively $\Gamma \leq 5/2$: at $r = 1, R = 6$ this is $3x^4 - 28x^3 + 66x^2 - 11 > 0$ on $[1, 6]$, whose endpoint values are $30,205$ and interior critical values $(85 \mp 35\sqrt{5})/2 > 0$. The integral increases with R and decreases with r.

On $\rho \in [l, u]$, $|\beta| \leq b$, $M_4 \leq M$, $|\gamma| \leq g$ and $0 \leq \rho - \gamma \leq \Delta$, the square $V = M_5 - 2\rho M_4 + \rho^3 \geq 0$ gives two complete constant bounds:

$$\begin{aligned} A_1(\Delta) &= 3b(2u+1)/2 + 3\Gamma(R, l)\Delta/2 + 3u^2(u+1)/8 + u^3/4 - l/2, \\ A_2 &= 3b(2u+1)/2 + 2uM - l/2. \end{aligned} \tag{30}$$

For $K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2$, intersect the applicable bounds

$$K_{\text{rad}} = 3ub + \frac{R(R+2-3l/2)}{R+1/2}(u-1), \quad K_{\text{univ}} = 6u - 3u^2/2 + 3b,$$
$$K_R = Ru/2 + \max_{r=l,u} |R/2 - 3r/2|g + 3b.$$

The first follows from $K_1 + 3\rho\beta = \int \operatorname{sgn}(X)x(x + 2 - 3\rho/2)/(x + 1/2)d\mu_0$. This ratio is convex in x, decreases in ρ, and its negative part is at most $31/40$ for $\rho \leq 1.85$; require the endpoint ceiling to exceed that number. The second uses $|\mathbb{E}[X|X|^3] - 3\gamma| \leq 3\rho$ for $R \leq 6$, $\rho \leq 2$; the third uses $0 \leq \mathbb{E}X_\pm^4 \leq R(\rho \pm \gamma)/2$. Finally $|\varphi\Omega| \leq v_H := \frac{2}{5}\max\{1/2, 2M/3 - 1/2\}$ for $\Omega = J_1 + (M_4 - 3)/6$.

Lemma 4.2 (Signed fourth-moment charge)**.** *Put $h_* = 1 + b$, $a(r) = (3r - R)/2$. Another bound for $|K_1|$ is $\max_{r\in[l,u]}\{Rr - 3r^2/2 + 3b + (1+b)a(r)^2/2\}$. If*

$$h_* \geq a(u), \quad 3 + 2a(l)h_* - 3h_*^2/2 \geq 0, \quad R - 3u + 3h_*^2/2 \geq 0,$$

intersect it with $Ru - 3u^2/2 - 3 + 3h_* + a(u)h_*^2 - h_*^3/2$.

Proof. For $\epsilon = \pm 1$, put $h = 1 - \epsilon\beta > 0$, $\delta = \rho - \epsilon\gamma$. Signwise Cauchy and $m \leq 1/2$ give $\delta \geq h^2$; fourth moments give

$$\epsilon K_1 \leq R\rho - 3\rho^2/2 - 3 + 3h + a(\rho)\delta - \delta^2/(2h).$$

Unrestricted square completion proves the first bound. Under $\delta \geq h^2$, the last three terms maximize to $h(3 + a^2/2)$ for $h \leq a$, and $3h + ah^2 - h^3/2$ for $h \geq a$. They agree at a. The first is increasing; the second has derivative $3 + 2ah - 3h^2/2$, nonnegative on $[0, h_*]$ by the second gate. The first gate selects its branch and the third makes its remaining ρ derivative nonnegative. Evaluate the first quadratic at endpoints and its eligible concave vertex. □

Call the applicable minimum K; no sign of $\beta\gamma$ is assumed.

Use 24 closed cells $[r_i, r_{i+1}]$, $r_i = 1 + 3361i/96000$, and

$$c_- = .4097321837023 < c < c_+ = .4097321837025,$$
$$\phi_l = 997/2500 < \phi(0), \qquad \sigma_l = 3.162277660 < \sqrt{10}.$$

The alternating Machin bounds and rational squares of Lemma B.1 license these constants. Put $\sigma_u = 3.162277661 > \sqrt{10}$. In pass 1 take $A = \min(A_1(2u), A_2)$, $f(t) = tK + v_H + s_F$, $q(t) = tl - g/5 > 0$. With overlines denoting upward rounding to 10^{-10}, set

$$\begin{aligned} T_1 &= \overline{\max_{t=c_-,c_+}\{K_F + tA + f(t)^2/(4q(t))\}}, \\ \eta_1 &= (c_- l - T_1/N)/2, \quad U_1 = \min\{T_1, \overline{\max_{t=c_-,c_+}[K_F + tA + f(t)^2/(4(q(t)+\eta_1))]}\}, \\ d_1 &= \overline{6U_1/(N\phi_l)}. \end{aligned} \tag{31}$$

Check $0 \le T_1 < Nc_- l$ before using η_1. The raw quadratic is affine in c at each $|H|$, so maximize at endpoints before completing squares. Corollary 2.3 then gives T_1, U_1, Nd_1. For pass 2 the moment circle gives $D \ge \sigma\rho - \gamma - 3\sqrt{\rho^2 - \gamma^2}$. Put $a_0 = \sigma_l - d_1/l$ and

$$y_l \le Y(a_0) := \begin{cases} -1, & a_0 \le -1, \\ (a_0 - 3\sqrt{10 - a_0^2})/10, & -1 < a_0 \le \sqrt{10}. \end{cases} \tag{32}$$

This smaller root inverts the increasing branch of $y + 3\sqrt{1-y^2}$ on $[-1, 1/\sqrt{10}]$; $a_0 - y \ge 0$ prevents the false squared branch. Round the root downward. Thus

$$\gamma \ge g_l := \max\{-g, \sigma_l l - 3 - d_1, \min(ly_l, uy_l)\},$$
$$\Delta = \min\{2u, (1-\sigma_l)l + 3 + d_1, u - g_l\}, \quad \phi_- = \phi_l(1 - T_1/(Nc_- l)),$$
$$A = \min\{A_1(2u), A_1(\Delta), A_2\}, \quad q(t) = \max\{tl - g/5, tl + \tfrac{1}{2}\min(\phi_- g_l, (2/5)g_l)\}.$$

Here $2m \le 1$ gives the additional bounds and both alternatives for q have slope l. Repeat the raw step, intersect with T_1, price phase only then, and intersect the resource with U_1 and d_2 with d_1.

Certificate condition 4.3 (Two coarse passes)**.** All coefficient, support, Fourier, denominator and strict raw gates above hold on all 24 cells, with outward ceilings

	$\max T$	$\max U$	$\max d$	strict gap gates
1	13.2818901598	10.9330610254	3.9164138937	23
2	11.5164967977	9.9790315907	3.5746638454	24

and $d_2 < (\sigma_l - 1)l$ in every second-pass cell, with minimum margin

$$\frac{526024602543}{1600000000000} > 0.$$

This is a finite exact obligation. Zero in the support would give $J \le \rho$ and $D \ge (\sigma - 1)\rho$, contradicting its gap test. Centering therefore acquires an actual central gap $(-A, B)$, $A, B > 0$.

4.4 Signed refinements of one physical law

Put $a_\pm = (1 \pm \beta)/2$, $b_\pm = (\rho \pm \gamma)/2$. Signwise Cauchy gives

$$m^2 \le a_+ a_-, \quad m b_\pm \ge a_\pm^2, \quad m \ge \max\{(1+\beta^2)/(2\rho), a_+^2/b_+, a_-^2/b_-\},$$
$$\frac{(1+\beta)^{3/2}}{\sqrt{1-\beta}} - \rho \le \gamma \le \rho - \frac{(1-\beta)^{3/2}}{\sqrt{1+\beta}}.$$

Both signs have positive mass and first moment, hence $|\beta| < 1$. Projection of $X|X|$ onto $(X, |X|)$, whose Gram matrix is $\left(\begin{smallmatrix} 1 & \beta \\ \beta & 1 \end{smallmatrix}\right)$, gives $M_4 \ge \rho^2 + (\gamma - \rho\beta)^2/(1-\beta^2)$. Integrating $(1, |X|)^{\mathsf{T}}(1, |X|)$ against $(R - |X|)dP$, $|X|(R-|X|)dP$ and $|X|^2(R-|X|)dP$ on each sign gives positive-semidefinite matrices, whose determinants imply

$$2m + \sum_\pm \frac{(Rm - a_\pm)^2}{D_\pm} \le R, \quad M_4 \le R\rho - \sum_\pm \frac{D_\pm^2}{Rm - a_\pm}, \quad D_\pm = Ra_\pm - b_\pm,$$
$$\mathbb{E}X_\pm^5 \le R\mathbb{E}X_\pm^4 - \frac{(Rb_\pm - \mathbb{E}X_\pm^4)^2}{D_\pm}.$$

Use corrections only at positive denominators. A positive numerator with zero denominator excludes only by its original determinant inequality; zero-over-zero corrections are dropped. The same m is retained on both signs. Integrating $(x-a)^2(R-x) \ge 0$ gives also $2m \le (R - \rho + 2a + Ra^2)/(a(2R+a))$; use $a = 2/10, \dots, 12/10$.

For $\delta = \gamma - \rho\beta$, three independent losses improve $\rho M_4/2 - M_5/4 \le \rho^3/4$:

$$P_1^* = \frac{(\rho+\gamma)\delta_+^2}{8(1+\beta)^2} + \frac{(\rho-\gamma)(-\delta)_+^2}{8(1-\beta)^2}, \quad P_2^* = \frac{\delta^4}{4\rho(1-\beta^2)^2}, \quad P_3^* = \frac{\delta^2}{8m}.$$

For the first, signwise Cauchy gives $c_4 \ge b_3^2/a_2$, $f_5 \ge c_4^2/b_3$, so $\rho c_4/2 - f_5/4 \le \rho^2 b_3/4 - (c_4 - \rho b_3)^2/(4b_3)$; minimize the square over that half-line. The second follows from $M_4^2 \le \rho M_5$ and the projection bound. For the third, $V = \mathbb{E}[|X|(|X|^2 - \rho|X|)^2]$ and weighted Cauchy give $\delta^2 \le 2mV$. Subtract their maximum, never their sum.

The physical decomposition (20) gives $T_+ + T_- \le D/(\sigma - 1)$ for $T_+ = b_+ - mB^2$, $T_- = b_- - mA^2$. Also $J \le \rho + 2mAB$ gives $6mAB \ge (\sigma-1)\rho - D$. On $\rho \ge l$, $D \le d < (\sigma_l - 1)l$, $m \le m_u$,

$$(A+B)^2 \ge \{l - d/(\sigma_l - 1) + ((\sigma_l - 1)l - d)/3\}/m_u > 0. \tag{33}$$

Retain $A \le a_-/m$, $B \le a_+/m$ and $D \ge \sigma\rho - \gamma - 6m$. In particular

$$m \ge m_* := \max\{0, (\sigma_l l - \gamma_+ - d)/6\}, \qquad |\beta| \le \sqrt{1 - 4m_*^2}. \tag{34}$$

An $m_* > 1/2$ excludes the cell. Intersect beta afresh in each gamma slice before recomputing dependent moments. The stronger outer-gap inequality gives $D \ge (\sigma-1)b_+ + 6a_+B - (\sigma + 5)mB^2$. If $D < (\sigma-1)b_+$, its concave quadratic forces B above the positive root; the analogue uses $A, a_-, b_-, \sigma+1, \sigma+7$. Round positive coefficients down, negative coefficients up and roots down.

The actual event $XX' < 0$ has probability $2pq$ and separation $A + B$. Cauchy gives $p \in [2m_-^2/(1+\beta_+), 1 - 2m_-^2/(1-\beta_-)]$ and $2pq \ge (1+\beta^2)^2/(2\rho^2)$. Exactly,

$$\mathbb{E}[(X - X')^2 \mid XX' < 0] = \frac{1 + 4m^2 + \beta}{2p} + \frac{1 + 4m^2 - \beta}{2q}.$$

This increases with m, is affine in β and convex in p, with positive numerators. Use its corner maximum or $2/\alpha$, whichever is smaller, with $\alpha = \max\{(1+\min|\beta|^2)^2/(2u^2), 2\min_{p=p_-,p_+} pq\}$. The downward root of (33) then licenses another complete HIGH option, minimized against the universal event.

For the signed integral charge put

$$P_1 = B^3/3 + AB^2/2 - A^3/6, \quad P_2 = B^4/4 + AB^3/3 - A^4/12, \quad G = \rho P_1 - P_2.$$

The central terms are $J_{1,c} = mP_1$, $J_{2,c} = mP_2$. Since $\partial_B G = B(A+B)(\rho - B)$,

$$G \le G(A, \rho; \rho) = \rho^4/12 + \rho^3 A/6 - \rho A^3/6 + A^4/12.$$

It increases with A (derivative $(\rho - A)^2(\rho + 2A)/6$) and is convex in ρ, so use A_u and both rho endpoints. Positive outer gaps satisfy $(t-a)(\rho - t) \le (\rho - B)_+^2/4$; negative outer gaps cost at most $(R-\rho)^2/4$ after extracting $|t|$. Integration gives

$$3(\rho J_1 - J_2)_o \le 3\{(\rho - B)_+^2 T_+ + (R-\rho)^2 T_-\}/8.$$

The joint budget is $D - mQ$, where $Q = (\sqrt{\sigma+1}A - \sqrt{\sigma-1}B)^2$; intersect it with $T_+ \le b_{+,u} - m_l B_l^2$, $T_- \le b_{-,u} - m_l A_l^2$. For Ω use $-RT_-/2 \le J_{1,o} \le RT_+/2$. The minimum of P_1 is at B_l and an A endpoint; the maximum is at B_u and A clipped to B_u. Multiply a nonnegative polynomial bound by m_u for an upper bound and m_l for a lower bound, reversing the choices for a negative bound. Fourth-semimoment intervals similarly enclose K_1. Retain the full signed pair (K_1, Ω).

4.5 Ordered update and closed coverage

Each cell carries its radius, moments, raw ceiling T and weighted invariant $nD \le Nd$. Intersect all preceding necessary conditions outwardly; monotone bisection for m retains its upper endpoint. Take

$$J_- = \max\{0, ((3+\sigma_l)l - \gamma_+ - d)/3\}, \quad J_+ = \min\{u + 2m_+, ((3+\sigma_u)u - \gamma_-)/3\}.$$

Let B_* be the corner maximum of $3\beta(J+\gamma)/2$, G_* the smaller coarse/central-plus-outer charge and P_* a lower bound for $\max(P_1^*, P_2^*, P_3^*)$. Then

$$A = \min\{B_* + G_* + u^3/4 - l/2 - P_*, B_* + 2uM - l/2\} \tag{35}$$

encloses the complete constant. Let $[k_-, k_+]$, $[\omega_-, \omega_+]$ enclose K_1, Ω. Under $0 \le T < Nc_-l$, put $\phi_- = \phi_l(1 - T/(Nc_-l)) > 0$ and

$$\begin{aligned} B_t &= K_F + tA, \quad F_t = \max_{k=k_\pm,\, \omega=\omega_\pm,\, \psi=\phi_-,2/5} |tk + \psi\omega| + s_F, \\ Q_t &= tl + \tfrac{1}{2}\min\{\phi_-\gamma_-, (2/5)\gamma_-\}. \end{aligned} \tag{36}$$

Thus $\mathcal{R}_c \le B_c + F_c|H| - Q_c H^2$. Successive affine maximization licenses the corners; convexity in c then licenses c_-, c_+ even when $A < 0$. Negative γ_- uses the upper density endpoint.

Lemma 4.4 (Ordered step)**.** *If $Q_t > 0$ at both endpoints, with all ceilings rounded upwards, set*

$$\begin{aligned} T' &= \min\{T, \max_{t=c_-,c_+}(B_t + F_t^2/(4Q_t))\}, \quad \eta = (c_-l - T'/N)/2, \\ U' &= \max_{t=c_-,c_+}(B_t + F_t^2/[4(Q_t+\eta)]), \quad d' = \min\{d, 6U'/(N\phi_l)\}. \end{aligned}$$

Negative T' or U' excludes the cell. Otherwise $\mathcal{R}_c \le T'$, $\mathcal{U} \le U'$ and $nD \le Nd'$, with

$$|H| \le \max_{t=c_-,c_+} \frac{F_t + \sqrt{F_t^2 + 4(Q_t+\eta)\max(0,B_t)}}{2(Q_t+\eta)}. \tag{37}$$

Proof. Complete the raw squares first. Since $T' \le T < Nc_-l$, phase pricing gives $\eta > 0$; only now complete the resource squares. The nonnegative resource contains $n\phi(0)D/6$, proving the weighted bound at actual n. A phase where both resource quadratics are negative is impossible; replacing B_t by its positive part enlarges the positive root and gives (37). Intersecting valid bounds preserves the claims; a resource ceiling cannot replace the raw ceiling in its license. □

First license each rho cell's radius by the heat/ray test. Each signed pass splits beta into 8 closed intervals, each necessary gamma range into 8, and the common m range into 2. Restore the original beta interval at every gamma slice before (34). A contradictory necessary condition excludes all descendants of that parent; active plus excluded equals 128 per processed rho cell. Pool upper bounds by maxima and lower bounds by minima; remove a rho cell only when every descendant is excluded. Closed seams may duplicate work, never lose a witness. For export retain all 100 closed panels $[j/100, (j+1)/100]$; a boundary law belongs to both incident panels. Use $\theta = 1 - m^2/(pq)$ with lower m and the full maximum of pq on each cell-panel intersection, including $1/4$ across $1/2$. Pool all bounds from those same cells.

Certificate condition 4.5 (Four signed passes)**.** Starting from Condition 4.3, all support, moment, actual-event, Fourier, denominator, signed-curvature and raw-license gates hold on this complete four-pass cover. Its 11264 nominal boxes and complete 100-panel export give, on every nonempty panel,

$$\mathcal{R}_c \le 2.1995174, \quad nD \le 32.20665378, \quad D \le .76682509,$$
$$\rho \le 71527/48000, \quad |\beta| \le .50549963, \quad |H| \le 1.83605680, \quad \operatorname{supp} P \subset (-157/50, 157/50).$$

Fourteen rho cells survive and 30 probability panels are nonempty.

Theorem 4.6 (Selected coarse entry)**.** *Under the two finite conditions, every attained pair with $a \ge 0$, $n \ge 42$ and true $C_{n-1} \le C_{\mathrm{E}}$ satisfies these bounds and the complete probability cover, assuming* (21)*.*

Proof. Support licenses higher moments, and the actual events and Fourier appendix bound the full integral. The two coarse passes acquire the gap. Every signed cell applies only necessary conditions of that same law and the ordered step. Induction over all four passes preserves the weighted invariants; closed coverage and outward pooling place every witness in an incident output panel. □

The numerical conditions remain exact finite obligations. The one-strip argument removes only downstream panels; this internal cover stays intact.

5 One acquired strip

The entry calculation will be used only through its weighted defect and raw-threshold bounds. The former already forces the sign split close to Esseen's two-point law. Two applications

of contact then bound the entire conditional support, without a further partition of the sign probability. Throughout this section set

$$\sigma = \sqrt{10}, \qquad p_* = (4-\sigma)/2, \qquad B_* = 32.20665378.$$

The law, support gaps and sign split are those of Section 3; in particular D is the actual gap defect, not a moment surrogate. Write the central gap, once acquired, as $(-A_-, A_+)$, so that its endpoints are not confused with the sign indicator $B_0 = \mathbf{1}_{\{X>0\}}$.

Proposition 5.1 (Direct geometry)**.** *Let X be compactly supported, centered and of variance one. If $n \geq 700$ and $nD \leq B_*$, its closed support has a gap $(-A_-, A_+)$ across zero, with $A_-, A_+ > 0$. In the actual decomposition $X = \ell(B_0 + U - p)$, put*

$$a_0 = p - A_-/\ell, \quad a_1 = q - A_+/\ell, \quad d = a_0 + a_1,$$
$$\theta = \mathbb{E}\{X - \mathbb{E}(X \mid B_0)\}^2, \qquad \lambda = n\mathbb{E}U^2 = n\theta/\ell^2.$$

Then $a_0, a_1 \geq 0$, $\mathbb{E}(U \mid B_0) = 0$, $U_0 \leq a_0$, $U_1 \geq -a_1$, and

$$\frac{3}{8} < p < \frac{93}{200}, \qquad \theta < .0036, \qquad d < .004, \qquad \ell > 1.996, \tag{38}$$
$$\rho < \frac{9}{8}, \qquad .066148 < \beta < .2527, \qquad \lambda < \frac{7}{20}. \tag{39}$$

These conclusions require no contact or predecessor hypothesis.

Proof. We have $D < D_0 := .047$, because $700 D_0 > B_*$. If zero were in the closed support, Lemma 3.1 would give $D \geq (\sigma-1)\rho \geq \sigma - 1 > D_0$. Thus the central gap exists; centering supplies both signs. Conditional means imply $a_j \geq 0$ and $0 \leq d < 1$. Put $v = pq$ and $w = 1 - \theta = v\ell^2$, and define

$$Q(p) = (\sigma+1)p^2 + (\sigma-1)q^2 - 6pq = (6+2\sigma)(p-p_*)^2.$$

We first retain all the costs of the actual conditional noises. Let

$$\int f \, d\mu_\pm = \mathbb{E}\big[(X - \mathbb{E}(X \mid B_0))^2 \mathbf{1}_{\{\pm X>0\}} f(|X|)\big].$$

Their masses sum to θ. The outer-gap estimate gives the common budget

$$D \geq \ell w Q + E_0 + \int k_+ \, d\mu_+ + \int k_- \, d\mu_-, \tag{40}$$

where

$$E_0 = 6\ell w\{d(1-d) + a_0 a_1\},$$
$$k_+(x) = (\sigma-1)(x + 2\ell q) + 6A_+, \qquad k_-(x) = (\sigma+1)(x + 2\ell p) + 6A_-.$$

Indeed

$$J_o^+ \leq 2\int_{A_+}^{\infty} (t - A_+)\tau(t)\, dt = \mathbb{E}[X_+(X_+ - A_+)^2],$$

and the negative outer gaps have the same expression in magnitudes. Add the central contribution $m(A_- + A_+)^2$ and use

$$\mathbb{E}X_+^3 = m(\ell q)^2 + \int (x + 2\ell q)\, d\mu_+, \qquad R_+ = \mu_+(\mathbb{R}) + m(\ell q - A_+),$$

with the negative counterparts. Expansion yields (40). This also proves the budget for countably many gaps by monotone convergence; endpoint atoms do not alter the integrals.

Hereafter use $3.162277660 < \sigma < 3.162277661$. The weaker gap estimate $J \le \rho + 2mA_-A_+$, together with $\rho \ge 1$, $\gamma \le \rho$ and $m \le 1/2$, gives

$$A_-A_+ \ge z := \frac{3.162277660 - 1 - D_0}{3} > 0, \qquad w \ge A_-A_+ \ge z.$$

The second inequality follows from $A_- \le \ell p$, $A_+ \le \ell q$. All costs in (40) are nonnegative, so

$$D \ge \frac{w^{3/2}Q}{\sqrt{v}} \ge 2z^{3/2}Q.$$

Let Q_- replace σ in its polynomial expression by the lower rational bound. The two exact tests

$$4z^3Q_-(.36)^2 > D_0^2, \qquad 4z^3Q_-(.48)^2 > D_0^2$$

exclude $p \le .36$ and $p \ge .48$, respectively: the squared distance to $p_* \in (.36, .48)$ is monotone outside that interval.

On $.36 < p < .48$, $\ell \ge 2\sqrt{z} > 1.67$, and the noise prices satisfy

$$k_+(x)/\ell \ge q\{(\sigma+5)z/w + 2(\sigma-1)\} > 5.2,$$
$$k_-(x)/\ell \ge p\{(\sigma+7)z/w + 2(\sigma+1)\} > 5.2.$$

For the first use $x \ge A_+$ and $A_+ \ge z/(\ell p)$; for the second use their negative counterparts. Since $w \le 1$, the constants follow by taking $q = .52$ and $p = .36$. Therefore $D \ge 1.67(5.2)\theta$, whence $\theta < .006$, $w > .994$ and $\ell > 1.99$. Also $A_- + A_+ \ge 2\sqrt{z}$ gives

$$d \le 1 - 2\sqrt{zv/w} < .2, \qquad v \ge .36(.64).$$

Evaluating the kernels at their lower support endpoints in (40) now yields

$$D \ge \ell\{wQ + 6wd(1-d) + \theta \min(C_+ - b_+d, C_- - b_-d)\}, \tag{41}$$

where

$$C_+ = 3(\sigma+1)q, \quad C_- = 3(\sigma+3)p, \quad b_+ = \sigma+5, \quad b_- = \sigma+7.$$

Each branch increases in d on the current domain, since $6(.994)(.6) - .006(3.162277661+7) > 0$; both noise prices are positive. The inward cost alone excludes $d \ge .004$, because

$$1.99(.994)6(.004)(.996) > .047.$$

After replacing d by zero, the derivative in θ of either branch $\sqrt{(1-\theta)/v}\{(1-\theta)Q + \theta C\}$ has the sign of $(2-3\theta)C - 3(1-\theta)Q$. It is positive: $C > 6.49$, $Q < .05$ and $\theta < .006$. Hence $D \ge Q/\sqrt{v}$. This ratio decreases on $[.36, 3/8]$, and

$$Q_-(3/8)^2 > D_0^2(3/8)(5/8)$$

excludes that interval. On $[93/200, .48]$ instead use $Q \ge Q(93/200) > D_0/2$ and $\sqrt{v} \le 1/2$. Thus $3/8 < p < 93/200$. Since

$$C \ge \min\{3(\sigma+1)(107/200),\ 3(\sigma+3)(3/8)\},$$

the bound $D \geq \ell\theta C$, with $\ell > 1.99$, improves $\theta < .0036$ and consequently $\ell > 1.996$.

The exact conditional cubic expansion gives

$$\rho = w^{3/2}\rho_B(p) + \int (x + 2\ell q)\, d\mu_+ + \int (x + 2\ell p)\, d\mu_-, \qquad \rho_B(p) = \frac{1 - 2v}{\sqrt{v}}.$$

Because the two integrands are priced by at least $\sigma - 1$ in (40),

$$\rho \leq \rho_B(p) + D/(\sigma - 1) < 1.1 + .047/(54/25) < 9/8.$$

Here $v \geq 15/64$ implies $\rho_B < 1.1$ by squaring positive quantities. The signed-second-moment identity gives

$$1 - 2p - 2q\theta \leq \beta \leq 1 - 2p + 2p\theta;$$

its endpoints imply $.066148 < \beta < .2527$.

To bound the scaled noise λ, retain the cubic cost rather than only its sign. If Z is centered, $Z \leq a$, $a > 0$, and $t = \mathbb{E}Z^2$, then

$$0 \leq \mathbb{E}[(a - Z)(Z + t/a)^2] = at - \mathbb{E}Z^3 - t^2/a.$$

Apply this to U_0 and $-U_1$. A zero endpoint forces the corresponding noise to vanish, so no division by zero is needed. For $\lambda_0 = nq\mathbb{E}U_0^2$, $\lambda_1 = np\mathbb{E}U_1^2$, the same gap-budget expansion, before bounding these two cubics, reads

$$\begin{aligned} nD/\ell^3 \geq\ & nvQ + 6nv\{d(1-d) + a_0a_1\} \\ & + \{3(\sigma+3)p - (\sigma+7)a_0\}\lambda_0 + \{3(\sigma+1)q - (\sigma+5)a_1\}\lambda_1 \\ & + \frac{(\sigma+1)\lambda_0^2}{nqa_0} + \frac{(\sigma-1)\lambda_1^2}{npa_1}. \end{aligned}$$

Terms with $a_j = 0$ are absent. Set $\bar{d} = .004$ and use $d(1-d) + a_0a_1 \geq (1-\bar{d})(a_0 + a_1)$. Pair each displacement cost with its reciprocal cubic cost by AM–GM; for example

$$6nv(1-\bar{d})a_0 + \frac{(\sigma+1)\lambda_0^2}{nqa_0} \geq 2\lambda_0\sqrt{6(1-\bar{d})(\sigma+1)p}.$$

Thus

$$nD \geq \ell^3\{nvQ + \lambda_0 Y_0(p) + \lambda_1 Y_1(p)\}, \tag{42}$$

where

$$\begin{aligned} Y_0 &= 3(\sigma+3)p - (\sigma+7)\bar{d} + 2\sqrt{6(1-\bar{d})(\sigma+1)p}, \\ Y_1 &= 3(\sigma+1)q - (\sigma+5)\bar{d} + 2\sqrt{6(1-\bar{d})(\sigma-1)q}. \end{aligned}$$

Each increases in its respective argument p or q. At $p = 3/8$ and $q = 107/200$, lower bounds 3 and 2.6 for the roots give $Y_0 > 12.8$, $Y_1 > 11.8$. Since $\lambda_0 + \lambda_1 = \lambda$,

$$B_* \geq nD > 1.996^3(11.8)\lambda \quad (\lambda > 0), \qquad 1.996^3(11.8)(.35) > B_*.$$

This proves $\lambda < .35$; at zero noise it is immediate. □

Proposition 5.2 (Two contact rays)**.** *Under Proposition 5.1, suppose (P, H) is an attained fixed-target pair at C_{E} with nonnegative excess and true predecessor $C_{n-1} \leq C_{\mathrm{E}}$. Then*

$$-.6 < U_0 \leq a_0, \qquad -a_1 \leq U_1 < .6.$$

The support conclusion is uniform in the raw threshold H.

Proof. Use the contact upper expression (22). The all-threshold Gaussian heat estimate and its radial derivative bound from the proof of Theorem 4.1 give, for $t > \sqrt{3}$,

$$H_n(H, \pm t) \le h(t) := S^3 \left\{ \frac{2}{5} \left(\frac{t^3 - 3t}{6} + \frac{|t^4 - 6t^2 + 3|}{72} \right) + \frac{t^5}{4100} + \frac{3t}{820} + \frac{17(3t^2 - 2)}{78720} \right\},$$
$$|\partial_t H_n(H, \pm t)| < g(t) := \frac{2}{5}(t^2/2 + 1/2 + \eta), \qquad S = 81/80, \quad \eta = 8/875.$$

The first bound follows directly from (23): $\|\phi'''\|_\infty < 2\phi(0)$, $n - 1 \ge 41$, and $\sqrt{n-1} > 6$. No derivative of the absolute-value majorant is taken.

At a support point of magnitude t, contact requires a nonnegative upper expression $H_n(H, \pm t) + C_{\mathrm{E}} Q_b(t)$, where

$$Q_b(t) = -t^3 + \tfrac{3}{2} r t^2 + 3bt + \eta r, \quad r = 9/8, \quad b = \begin{cases} -.066148 & \text{negative ray,} \\ .2527 & \text{positive ray.} \end{cases}$$

These replacements follow with their signs from (39); the discarded excess term is nonpositive. Take $c_0 = 819/2000 < C_{\mathrm{E}}$ and the two fixed boundaries

$$t_- = 1.996(3/8 + .6) = 1.9461, \qquad t_+ = 1.996(107/200 + .6) = 2.26546.$$

For the respective b, direct rational substitution at each boundary gives

$$t^2 > 3, \qquad h(t) + c_0 Q_b(t) < 0, \qquad Q_b, Q_b', Q_b'' < 0, \qquad L, L', L'' < 0, \quad L = g + c_0 Q_b'.$$

The two boundary upper bounds are less than $-.40$ and $-.13$, respectively. These are just polynomial comparisons at two rational points. Since $Q_b''' = -6$ and $L'' = 2/5 - 6c_0 < 0$, the required signs persist along both outward rays. The derivative of the actual upper expression is below L, as $Q_b' < 0$ and $C_{\mathrm{E}} > c_0$. It therefore stays strictly negative beyond either boundary. Contact excludes every support point there, including boundary atoms.

On the negative class, $X > -t_-$ and $\ell > 1.996$ give $U_0 = X/\ell + p > -(3/8 + .6) + p > -.6$. On the positive class, $X < t_+$ gives $U_1 = X/\ell + p - 1 < 107/200 + .6 + p - 1 < .6$. The inward endpoints and conditional centering are the actual ones from Proposition 5.1. □

The pointwise normalization is worth recording once:

$$1 = v\ell^2 + \theta, \qquad s = \mathbb{E}U^2 = \theta/\ell^2, \qquad v + s = \ell^{-2}, \qquad X = \frac{B_0 + U - p}{\sqrt{v + s}}. \tag{43}$$

Thus the law to be compared in the next section is exactly the original law, and its threshold remains the raw H. Reflection, when required, selects the oriented positive witness before acquisition; neither its signed defect budget nor its probability interval is transported from the opposite orientation.

6 The direct comparison with a binomial law

The acquired strip admits a single comparison on its whole probability interval. Very small noise is controlled by the loss at one count and fourth moments. For the rest, five adjacent counts give a uniform crossing loss. Both arguments concern the actual conditional laws of one marginal.

Theorem 6.1 (The common-strip comparison)**.** *Let $n \ge 700$ be an integer, $B \sim \mathrm{Bernoulli}(p)$, and $\mathbb{E}(U \mid B) = 0$, where $p \in [3/8, 93/200]$. Write U_j for the conditional noise given $B = j$, and suppose*

$$-3/5 \le U_0 \le a_0, \qquad -a_1 \le U_1 \le 3/5, \qquad a_0, a_1 \ge 0, \quad a_0 + a_1 \le 1/250.$$

Put $q = 1 - p$, $v = pq$, $s = \mathbb{E}U^2$, $\lambda = ns$, $X = (B + U - p)/\sqrt{v + s}$ and $\rho = \mathbb{E}|X|^3$. For independent copies and every raw threshold $|H| \le 46/25$,

$$\sqrt{n}\left\{\mathbb{P}\left(\sum_{i=1}^{n} X_i \le H\right) - \Phi(H/\sqrt{n})\right\} - C_{\mathrm{E}}\rho < -\lambda/100 \qquad (0 < \lambda \le 7/20). \tag{44}$$

At $\lambda = 0$ the same excess is strictly negative.

The proof below uses only the displayed hypotheses. Contact and the sharp predecessor are needed earlier to acquire this strip, not for its local comparison. The small- and larger-noise intervals meet at $\lambda = 1/100$.

6.1 Three elementary ingredients

Write $\pi_j = \binom{n}{j} p^j q^{n-j}$, $F_B(j) = \sum_{i \le j} \pi_i$, and $\rho_B = (1 - 2v)/\sqrt{v}$. We use Schulz's strict finite-binomial theorem [4, Theorem 1] in the form

$$\sqrt{n}\{F_B(j) - \Phi((j - np)/\sqrt{nv})\} < C_{\mathrm{E}}\rho_B \quad (0 < p < 1, n < \infty). \tag{45}$$

Its strictness concerns every finite binomial law; it is not inferred from the limiting lower approximation for the universal constant.

The factorial estimates used throughout are finite Stirling bounds:

$$\log(j!) = (j + \tfrac{1}{2})\log j - j + \tfrac{1}{2}\log(2\pi) + r_j, \qquad 0 < r_j < 1/(12j).$$

For completeness, with $x = 1/(2j+1)$, $r_j - r_{j+1} = \operatorname{arctanh}(x)/x - 1 = \sum_{l \ge 1} x^{2l}/(2l+1)$ lies strictly between 0 and $1/(12j(j+1))$. Wallis's product fixes the limiting constant, so $r_j \to 0$; summing these differences proves the bounds. One useful consequence is

$$\max_j \mathbb{P}\{\mathrm{Bin}(M, p) = j\} < \frac{41}{100\sqrt{Mv}} \quad (M \ge 43, p \in [1/3, 2/3]). \tag{46}$$

Indeed a mode has $z = j/M = p + \delta/M$, $|\delta| \le 2/3$. Since $v \ge 2/9$ and $|1 - 2p| \le 1/3$, $z(1 - z) \ge v(1 - 1/M - 2/M^2) > 39v/40$. Stirling, the nonnegative relative entropy, and $e^{1/516} < 516/515$ bound the mass by $(2/5)\sqrt{40/39}(516/515)/\sqrt{Mv}$, which is smaller than the right side of (46) by a rational square comparison.

Finally the cubic correction has the favorable sign on the full strip. Let $s_j = \mathbb{E}U_j^2$, so $s = qs_0 + ps_1$, and set $r = 3/8$, $d = 1/250$, $v_* = 15/64$. The conditional supports have fixed signs relative to p, since $d < r$. Expansion of the raw third absolute moment gives

$$\mathbb{E}|B + U - p|^3 = v(1 - 2v) + 3v(s_0 + s_1) - q\mathbb{E}U_0^3 + p\mathbb{E}U_1^3.$$

The inward bounds imply $\mathbb{E}U_0^3 \le a_0 s_0$, $\mathbb{E}U_1^3 \ge -a_1 s_1$. Together with $(1 + x)^{3/2} \le 1 + 3x/2 + 3x^2/8$ for $x \ge 0$, this yields

$$\begin{aligned}(v + s)^{3/2}(\rho - \rho_B) &\ge s\left\{3r - d - \frac{3}{2}(1 - 2v) - \frac{3(1 - 2v)s}{8v}\right\} \\ &\ge s\left\{\frac{2593}{8000} - \frac{17}{20}s\right\} > 0, \qquad 0 < s \le \frac{1}{2000}.\end{aligned} \tag{47}$$

Here $s \le (7/20)/700 = 1/2000$, and the displayed coefficient is $3r - d - \frac{3}{2}(1 - 2v_*)$. Thus every nonnegative coefficient of $\rho_B - \rho$ can be discarded in an upper comparison.

6.2 Small noise: one count and the whole incoming tail

Assume $0 < \lambda \le L = 1/100$. Choose an integer k with

$$t = np + H\sqrt{v+s} = k + u, \qquad -3/10 \le u \le 7/10, \qquad \delta = k - np.$$

The phase interval has length one. It need not use the same anchor as the larger-noise proof. Let W_j have the law of the sum of j independent U_1 variables and $n-j$ independent U_0 variables. For $C \ge 0$, subtract the signed binomial excess at k from the actual excess:

$$\Delta_C = \sqrt{n}\left\{\mathbb{P}\left(\sum X_i \le H\right) - \Phi(H/\sqrt{n})\right\} - C\rho \\ - \sqrt{n}\{F_B(k) - \Phi(\delta/\sqrt{nv})\} + C\rho_B.$$

Exact conditioning gives

$$\mathbb{P}\left(\sum X_i \le H\right) - F_B(k) = -\sum_{i=0}^{k} \pi_{k-i}\mathbb{P}(W_{k-i} > u+i) \\ + \sum_{i=1}^{n-k} \pi_{k+i}\mathbb{P}(W_{k+i} \le u-i).$$

Outgoing events are strict and incoming events closed; this convention includes all threshold atoms. Define

$$f(u) = \frac{\Phi((\delta+u)/\sqrt{n(v+s)}) - \Phi(\delta/\sqrt{n(v+s)})}{\pi_k}, \qquad O = \mathbb{P}(W_k > u) + f(u),$$

$$T_{\mathrm{in}} = \sum_{i=1}^{n-k} \mathbb{P}(W_{k+i} \le u-i), \qquad \mathcal{R} = \sqrt{n}\{\Phi(\delta/\sqrt{nv}) - \Phi(\delta/\sqrt{n(v+s)})\} + C(\rho_B - \rho).$$

Keeping the own-count loss and discarding the other nonpositive outgoing terms gives

$$\Delta_C \le -m_k(O - RT_{\mathrm{in}}) + \mathcal{R}, \qquad m_k = \sqrt{n}\pi_k, \tag{48}$$

whenever $\pi_{k+i}/\pi_k \le R$ for all legal incoming counts.

Uniform normalization.

Set $D_* = 5/3$, $b_* = 1/20$, $T_* = 101/10000$ and $c_* = 2/525$. Because $v \ge v_*$ and $v + s < 1/4$, $|t - np| < 23/25$. Consequently $|\delta| < D_*$ globally and $|\delta| < 1$ when $|u| \le b_*$. Write $\mathcal{K}(z,p) = z\log(z/p) + (1-z)\log((1-z)/q)$ for binomial relative entropy; this is distinct from the physical gap defect D. The segment between p and k/n lies inside $[.37, .47]$; in particular $1 \le k \le n-1$. On this segment

$$z(1-z) > 23/100, \quad |\partial_z^3 \mathcal{K}(z,p)| < 5, \quad |(\log(z(1-z)))'| < 6/5, \quad (12z(1-z))^{-1} < 3/8.$$

Stirling and entropy Taylor expansion therefore give

$$m_k = \frac{\phi(0)}{\sqrt{v}} e^{-\delta^2/(2nv)+e}, \qquad n|e| < Z(D_0) := 3D_0/5 + 3/8 + D_0^3/840 \tag{49}$$

for $|\delta| \le D_0 \le D_*$. The three terms respectively bound the logarithmic prefactor, factorial remainder and entropy remainder. Also

$$\log f'(z) = -\tfrac{1}{2}\log(1 + s/v) + \frac{\delta^2}{2nv} - \frac{(\delta+z)^2}{2n(v+s)} - e,$$

so, on an integration interval $|z| \le b_0$,

$$n|\log f'(z)| \le \frac{11}{5}\lambda + \frac{2D_0 b_0 + b_0^2}{2v_*} + Z(D_0).$$

The coefficient $11/5$ exceeds $1/(2v_*) + D_*^2/(1400v_*^2)$. Using $e^{-x} \ge 1-x$, $e^x \le (1-x)^{-1}$ and $\phi(0) > 199/500$ now gives

	$\lvert u\rvert \le 1/20$	$-3/10 \le u \le 7/10$
m_k lower bound	$3/4$	$3/4$
f' lower bound	$99/100$	$49/50$
f' upper bound	$101/100$	$51/50$
R upper bound	$51/50$	$6/5$

(50)

The derivative estimates hold on the whole segment from 0 to u. For the global ratio, (46) gives $\sqrt{n}\max_j \pi_j < 17/20$, hence $R < (17/20)/(3/4) < 6/5$. In the inner interval the first ratio is

$$\frac{\pi_{k+1}}{\pi_k} = 1 + \frac{p - \delta - 1}{nv + q(\delta+1)} < 1 + \frac{93/200}{700 v_*} < 51/50;$$

every later ratio is at most one, since $p - \delta < 2$. By (47) the moment correction is nonpositive. Differentiating the normal CDF and using $\phi(0)/(2v_*^{3/2}) < 2$ therefore gives

$$\mathcal{R} < 2\lambda/n \quad (|u| \le b_*), \qquad \mathcal{R} < 4\lambda/n \quad \text{on the full phase interval.} \tag{51}$$

Indeed the positive displacement is bounded by 1 or $D_* < 2$.

No truncation of the incoming tail.

Since $|s_1 - s_0| \le (8/3)s$, the actual own-count variance $t_0 = \mathbb{E}W_k^2$ satisfies $.99\lambda < t_0 < 1.01\lambda \le T_*$. For each incoming count,

$$t_i = \mathbb{E}W_{k+i}^2 \le \lambda\{1 + c_*(D_0 + i)\}, \qquad \mathbb{E}W_{k+i}^4 \le A^2 t_i + 3t_i^2, \quad A = 3/5,$$

where $D_0 = 1$ in the inner interval and $D_0 = D_*$ otherwise. The moment bound follows from independence and $|U_j| \le A$. For $a = 1 + c_* D_0$ put

$$K(a) = A^2\left(\frac{13a}{12} + \frac{5c_*}{4}\right) + 3L\left(\frac{13a^2}{12} + \frac{5ac_*}{2} + \frac{5c_*^2}{3}\right).$$

Fourth-moment Markov at distance $i - u$, and extension of the nonnegative sum to all positive integers, give

$$\begin{aligned} T_{\text{in}} &< \frac{K(1+c_*)}{(1-b_*)^4}\lambda < .53\lambda \quad (|u| \le b_*), \\ T_{\text{in}} &< \frac{.43\lambda}{(1-u)^4} \qquad\qquad (0 \le u \le .7), \\ T_{\text{in}} &< .43\lambda \qquad\qquad (u \le 0). \end{aligned} \tag{52}$$

Here $K(1 + c_* D_*) < .43$, and $\sum i^{-4} < 13/12$, $\sum i^{-3} < 5/4$, $\sum i^{-2} < 5/3$. These series bounds follow by retaining respectively five, three and five terms and estimating the remaining sum by an integral. For $u \ge 0$ use $i - u \ge (1-u)i$; for $u \le 0$ use $i - u \ge i$.

The own-count loss at the central phases.

With $\alpha = 13/25$ and $M = 5\alpha^4$, the quartic

$$h_\alpha(x) = \frac{-x^4 + 3\alpha^2 x^2 + 2\alpha^3 x}{M} \le \mathbf{1}_{(0,\infty)}(x)$$

is a strict-event minorant: its numerator factors as $-x(x+\alpha)^2(x-2\alpha)$, and its positive maximum divided by M is $(9+6\sqrt{3})/20 < 1$. Let $r = zt_0$ and set

$$F_{\varepsilon,b}(z,t) = \frac{3\alpha^2 - A^2 - 3t}{M} + \left(\frac{2\varepsilon}{5\alpha} + b - \frac{4At}{M}\right) z + \frac{3\alpha^2 t - 6t^2}{M} z^2 - \frac{t^3 z^4}{M}.$$

Expanding $h_\alpha(W_k + r)$ for $u = -r$, or $h_\alpha(W_k - r)$ for $u = r$, and using $|\mathbb{E}W_k^3| \le At_0$, $\mathbb{E}W_k^4 \le A^2 t_0 + 3t_0^2$, yields

$$O/t_0 \ge F_{1,-101/100}(z,t_0) \quad (u=-r), \qquad O/t_0 \ge F_{-1,99/100}(z,t_0) \quad (u=r).$$

Both functions exceed $3/5$ on $0 \le z \le 3/2$, $0 \le t \le T_*$. To check the whole rectangle, use concavity in t and check only $t = 0, T_*$. At those endpoints replace z^4 by $(9/4)z^2$ in its negative term. The negative-phase quadratic decreases on $[0, 3/2]$, and the positive-phase quadratic increases; their minima are respectively at $z = 3/2$ and $z = 0$. All four rational endpoint values exceed $3/5$.

For negative phases $3t_0/2 \le r \le b_*$, Cantelli gives

$$O \ge \frac{r^2}{t_0 + r^2} - \frac{101}{100} r > \frac{3}{5} t_0.$$

Its derivative at $r = 3t_0/2$ is positive and first increases, then decreases. The minimum is therefore at an endpoint. At the first use $t_0 \le T_*$ after division by t_0; at the other use $b_*^2/(T_* + b_*^2) - (101/100)b_* > (3/5)T_*$. For positive phases beyond this switch, $O \ge (99/100)r \ge (297/200)t_0 > (3/5)t_0$. The switch lies below b_* because $3T_*/2 < b_*$. Thus

$$O > \frac{3}{5} t_0 > .59\lambda \qquad (|u| \le b_*). \tag{53}$$

The value $u = 0$ is included, with $h_\alpha(0) = 0$.

The outer phases and the final margin.

For $u = -r$, $b_* \le r \le 3/10$, Cantelli gives

$$O \ge \frac{r^2}{T_* + r^2} - \frac{51}{50} r > 7/50 \ge 14\lambda.$$

The derivative is positive at b_* and has at most one downward zero; both endpoint values exceed $7/50$. For $b_* \le u \le 7/10$,

$$O - RT_{\text{in}} \ge \frac{49}{50} u - \frac{(6/5)(43/100)L}{(1-u)^4} > 1/25 \ge 4\lambda.$$

This function is concave and both endpoints exceed $1/25$. In the inner interval, (52) and (53) give $O - RT_{\text{in}} > (.59 - (51/50).53)\lambda$. Substitute these three margins and (51) into (48), using $m_k > 3/4$. The smallest final margin is the inner one, and

$$\frac{3}{4}\left(\frac{59}{100} - \frac{51}{50}\frac{53}{100}\right) - \frac{2}{700} > \frac{1}{100}.$$

Hence $\Delta_C < -\lambda/100$. Taking $C = C_{\text{E}}$ and applying (45) proves (44) for $0 < \lambda \le 1/100$.

6.3 Larger noise: five counts and four endpoint checks

Now assume $1/100 \le \lambda \le 7/20$ and choose $k = \lfloor t \rfloor$, $u = t - k \in [0,1)$, $w = t - np$. The following constants will be used only in this subsection:

$$N = 700, \quad p_- = 3/8, \quad p_+ = 93/200,$$
$$r = 3/8, \quad q_* = 107/200,$$
$$v_- = 15/64, \quad v_+ = 9951/40000, \quad A = 3/5, \quad d = 1/250,$$
$$\eta = 2/5, \quad \beta_0 = 41/100, \quad \pi_+ = 22/7, \quad \phi_- = 199/500, \quad \phi_+ = 2/5,$$
$$\kappa = 3, \quad \xi = 93/100, \quad c_- = 6, \quad \zeta = 23/100, \quad m_* = 3/4, \quad V_* = 113/1000,$$
$$\bar{c} = 79/200, \quad j_0 = 1/5, \quad j_1 = 5/2.$$

In particular $|w| < \xi < 1$, since $(46/25)^2(v_+ + (7/20)/N) < \xi^2$. Also $N(r-d) > \xi$, so $na_0 \le t < n(1-a_1)$.

An exact weighted crossing identity.

On an auxiliary space take independent copies of the conditional laws and write

$$T_j = j + \sum_{l \le j} U_{1l} + \sum_{l > j} U_{0l}, \qquad L_j(t) = \mathbb{E}(t - T_j)_+, \qquad V_i = \sum_{l < i} U_{1l} + \sum_{l > i} U_{0l}.$$

The mesh is increasing because $T_j - T_{j-1} \ge 1 - a_0 - a_1 > 0$. Each T_j has the conditional convolution given j successes. Set

$$h(x) = \mathbb{E}(x - U_0)_+ - \mathbb{E}(x - 1 - U_1)_+ - \mathbb{P}(1 + U_1 \le x), \qquad g_i = \mathbb{E}h(t - i + 1 - V_i).$$

Independence of the omitted coordinate gives $g_i = L_{i-1}(t) - L_i(t) - \mathbb{P}(T_i \le t)$. Here $L_0 = t$, $L_n = 0$ and $\mathbb{P}(T_0 \le t) = 1$. Multiplication by π_i and finite summation by parts yield

$$\mathbb{P}\left(\sum X_i \le H\right) = (1-u)F_B(k) + uF_B(k+1) + \sum_{j=1}^{n-1}(\pi_{j+1} - \pi_j)B_j - \sum_{i=1}^{n} \pi_i g_i, \quad B_j = L_j(t) - (t-j)_+ \ge 0. \tag{54}$$

Indeed replacing L_j by $(t-j)_+$ in $\pi_0 + \pi_1 t + \sum(\pi_{j+1} - \pi_j)L_j$ telescopes to the stated interpolation. All CDF events are closed, including atoms.

Five kernels with four centered replacements.

Let $f(x) = \sin^2(\pi x)$ on $[0,1]$ and zero outside. It is continuously differentiable with derivative Lipschitz constant $2\pi^2$. We claim

$$h(x) \ge \eta f(x) \ge 0. \tag{55}$$

For $x < 1 - a_1$, $h(x) = \mathbb{E}(x - U_0)_+ \ge 0$, and Jensen gives $h(x) \ge x$ when $x \ge 0$. On $[0,1]$, $f(x) \le (5/2)x$: use $\pi^2 < 10$ up to $1/4$, the concave tangent $1/2 + \pi(x - 1/4)$ on $[1/4, 2/5]$, and $f \le 1$ thereafter. For $x \ge 1 - a_1 > a_0$, put $D_1 = 1 + U_1$ and $L_1 = \mathbb{E}(D_1 - x)_+$. Put–call parity gives $h(x) = \mathbb{P}(D_1 > x) - L_1 \ge 0$, since $D_1 \le 1 + A$ and $x > A$. When $1 - a_1 \le x \le 1$, $L_1 \ge 1 - x$ and

$$h(x) \ge \frac{x - A}{1 + A - x} L_1 \ge \frac{1 - A - d}{A + d}(1 - x) \ge \eta\pi^2(1-x)^2 \ge \eta f(x).$$

The penultimate inequality uses $\eta\pi_+^2 d \le (1-A-d)/(A+d)$. At $x=0,1$ both comparisons are weak, as required for atoms.

Freeze $V = V_{k+1}$ and let $V^{(j)}$ have the law of V_{k+1+j}. For $j>0$ it replaces j copies of U_0 by U_1. With $f_j(x) = f(u-j-x)$, regroup the two positive changes as

$$\begin{aligned}&\mathbb{E}f_1(V^{(1)}) + \mathbb{E}f_2(V^{(2)}) - \mathbb{E}f_1(V) - \mathbb{E}f_2(V)\\ &\quad = \mathbb{E}(f_1+f_2)(V^{(1)}) - \mathbb{E}(f_1+f_2)(V) + \mathbb{E}f_2(V^{(2)}) - \mathbb{E}f_2(V^{(1)}).\end{aligned}$$

There is an identical grouping for the negative changes. Adjacent bumps have disjoint open supports and zero derivatives at the joins, so the derivative Lipschitz constant remains $2\pi^2$. For a centered Z, $|\mathbb{E}F(a+Z)-F(a)| \le \frac{1}{2}\operatorname{Lip}(F')\mathbb{E}Z^2$. Conditioning on all unchanged summands, and applying this twice to each replacement, bounds the total cost by $4\pi^2(s_0+s_1)$. Hence

$$\sum_{i=k-1}^{k+3} g_i \ge \eta\left\{\sum_{j=-2}^{2}\mathbb{E}f(u-j-V) - 4\pi^2(s_0+s_1)\right\}.$$

These five indices are interior, because $|i-np| < \xi+3 < 4$. No independence between different count sums is asserted.

The infinite translates of f sum to $\sin^2(\pi(u-V))$. An omitted nonzero translate requires $|V|>2$, including endpoint collisions because the bump vanishes there. Thus the omitted part is at most $\mathbb{E}V^4/16$. With $\delta = k-np \in (-1-\xi, \xi)$, the frozen variance is

$$\tau = (nq-\delta-1)s_0 + (np+\delta)s_1, \qquad (1-6/n)\lambda \le \tau \le (1+3/n)\lambda.$$

The lower estimate uses $c_- r \ge 1+\xi$; the upper uses $r \ge 1/3$. Also $s_0+s_1 \le \lambda/(nr)$ and $\mathbb{E}V^4 \le A^2\tau + 3\tau^2$. For either conditional noise Z, an independent copy satisfies $|Z-Z'| \le A+d < 2/3$. Concavity of sine and $\sin(2\pi/3) = \sqrt{3}/2 > 17/20$ give $\sin(\pi x) \ge (51/40)x$ on $[0, A+d]$; hence $1-\cos(2\pi x) \ge 3x^2$ on $|x| \le A+d$. Therefore

$$1 - |\mathbb{E}e^{2\pi i Z}|^2 = \mathbb{E}\{1-\cos(2\pi(Z-Z'))\} \ge 2\kappa \operatorname{Var} Z,$$

and $|\mathbb{E}e^{2\pi i Z}| \le e^{-\kappa \operatorname{Var} Z}$. Multiplying over the independent frozen summands shows $\mathbb{E}\sin^2(\pi(u-V)) \ge (1-e^{-\kappa\tau})/2$. Altogether,

$$\sum_{i=k-1}^{k+3} g_i \ge \eta\left\{\frac{1-e^{-\kappa(1-6/n)\lambda}}{2} - \frac{A^2(1+3/n)\lambda + 3(1+3/n)^2\lambda^2}{16} - \frac{4\pi^2\lambda}{nr}\right\}. \tag{56}$$

The weights and the complete curvature tail.

For $k-1 \le i \le k+3$, the segment between i/n and p has variance product greater than ζ; indeed $\zeta < \min_{z=p_- -4/N,\, p_+ +4/N} z(1-z)$. Relative entropy is bounded by $(i-np)^2/(2n\zeta) < 8/(n\zeta)$, and the factorial remainder is greater than $-1/(12n\zeta)$. Stirling therefore gives

$$\sqrt{n}\pi_i > 2\phi_-\left(1 - \frac{8+1/12}{N\zeta}\right) > m_*.$$

Every other g_i is nonnegative. The reserve in (56) increases with n. Replace n by N first, then use $1-e^{-x} \ge x - x^2/2 + x^3/6 - x^4/24$ for $x \ge 0$. Writing $\alpha = 1-6/N$, $b = 1+3/N$ and $z = \kappa\alpha$, we obtain

$$\sqrt{n}\sum_i \pi_i g_i \ge W(\lambda) := \eta m_*\left\{\frac{z\lambda - z^2\lambda^2/2 + z^3\lambda^3/6 - z^4\lambda^4/24}{2} - \frac{A^2 b\lambda + 3b^2\lambda^2}{16} - \frac{4\pi_+^2\lambda}{Nr}\right\}. \tag{57}$$

Put $c = (np - q - t)_+$. For $a = t - j$ and $W_j = T_j - j$, centering gives

$$B_j = (\mathbb{E}|a - W_j| - |a|)/2 \geq 0, \qquad B_j \leq \sqrt{T}/2, \qquad a_+ B_j \leq T/4,$$

whenever $T \geq \mathbb{E}W_j^2$. The factorization $27x^4 - 256x + 256 = (3x-4)^2(3x^2+8x+16)$ gives $B_j \leq 27M/(256|a|^3)$ for $a \neq 0$, if $M \geq \mathbb{E}W_j^4$. The exact adjacent-mass formula is

$$\pi_{j+1} - \pi_j = \frac{np - q - j}{(n+1)v}\,\mathbb{P}\{\mathrm{Bin}(n+1, p) = j+1\}.$$

Discard its nonpositive terms and bound the positive numerator by $(a+c)_+$. Since $c \leq \bar{c} < 1$, at most one mesh point lies in each of $(-c, 0)$ and $[0, 1)$; together they cost $c\sqrt{T} + T/4$. The rest is bounded by the complete series $\sum i^{-2} < 5/3$, $\sum i^{-3} < 5/4$. Using (46) and $\sqrt{n}/(n+1)^{3/2} \leq 1/n$ yields

$$\sqrt{n}\sum_j (\pi_{j+1} - \pi_j) B_j \leq \frac{\beta_0}{nv^{3/2}} \left\{ \frac{T}{4} + c\sqrt{T} + \frac{27M}{256}\left(\frac{5}{3} + \frac{5c}{4}\right) \right\}. \tag{58}$$

A positive adjacent mass requires $j < np$, where its actual variance is at most $\lambda/q \leq T = \lambda/q_*$ and $\mathbb{E}W_j^4 \leq M = A^2T + 3T^2$. Moreover $J(\lambda) = j_1\lambda + j_0 \geq \sqrt{T}$, since $4j_0j_1 \geq 1/q_*$.

Both signs of the normal correction.

Apply (45) at $k, k+1$ and interpolate. For $G(x) = \Phi((x - np)/\sqrt{nv})$, $G''(x) \leq \phi(0)(np - x)_+/(nv)^{3/2}$. The interpolation remainder is

$$\sqrt{n}\{(1-u)G(k) + uG(k+1) - G(t)\} \leq \frac{\phi(0)}{2nv^{3/2}} u(1-u)(u-w)_+.$$

If $H \leq 0$, changing normal variance from v to $v + s$ is favorable. Using $u(1-u) \leq 1/4$ and $u^2(1-u) \leq 4/27$, the total normal cost is at most $\phi_+(\xi/8 + 2/27)/(nv^{3/2})$, and $c = (-q - w)_+ \leq \bar{c}$. If $H \geq 0$, $c = 0$, interpolation costs at most $\phi_+(2/27)/(nv^{3/2})$, and the variance change costs at most $\phi(0)ws/(2v^{3/2})$ by $1 - (1+x)^{-1/2} \leq x/2$. The cubic correction is favorable by (47).

Define the nonnegative quadratics

$$S_c = \frac{T}{4} + cJ(\lambda) + \frac{27M}{256}\left(\frac{5}{3} + \frac{5c}{4}\right), \qquad E_- = \frac{\beta_0 S_{\bar{c}} + \phi_+(\xi/8 + 2/27)}{V_*},$$

$$E_+ = \frac{\beta_0 S_0 + \phi_+(2/27 + \xi\lambda/2)}{V_*}.$$

Because $V_*^2 < v_-^3$, combining (54)–(58) with these normal bounds makes the full excess strictly less than $-W(\lambda) + E_\pm(\lambda)/n$. It remains to compare these explicit polynomials.

Four rational endpoints finish the interval.

Set $P_\pm(\lambda) = W(\lambda) - E_\pm(\lambda)/N - \lambda/100$. Each $E_\pm$ is a convex quadratic, whereas direct differentiation gives

$$W''(\lambda) = -\eta m_* \left\{ \frac{z^2}{4}\left[(z\lambda - 1)^2 + 1\right] + \frac{3b^2}{8} \right\} < 0.$$

Thus $P_\pm$ is strictly concave. Substitution of the rational constants above gives the following exact rational lower bounds:

	$\lambda = 1/100$	$\lambda = 7/20$
P_-	.002314	.061304
P_+	.003341	.063680

All four are positive, proving $P_\pm > 0$ on the whole closed interval. Since $n \geq N$ and $E_\pm \geq 0$, this proves (44) for $1/100 \leq \lambda \leq 7/20$. The scalar checks used here involve only rational inequalities and $199/500 < \phi(0) < 2/5$, $25/8 < \pi < 22/7$; no parameter grid or tail truncation is involved.

The two positive-noise proofs have identical support and probability hypotheses, include their shared endpoint, and cover every phase. At $\lambda = 0$, conditional centering and $\mathbb{E}U^2 = 0$ give $U = 0$ almost surely, so (45) applies directly. This completes the proof of Theorem 6.1.

7 Composition and the finite boundary

Theorem 7.1 (Acquired-witness exclusion)**.** *Let $n \geq 700$ and let (P, H) attain the fixed-target maximum with $a \geq 0$. Assume $C_{n-1} \leq C_{\mathrm{E}}$, the structural inequality* (21)*, and complete entry Conditions 4.3 and 4.5. Then such a pair cannot exist.*

Proof. Entry gives $nD \leq 32.20665378$ and $|H| \leq 1.83605680$ for the selected oriented law. Section 5 gives $3/8 < p < 93/200$, inward sum below $1/250$, amplitude $3/5$ and $\lambda < 7/20$. By (43), the standardized Bernoulli-plus-noise marginal is exactly P, with the same raw H. Theorem 6.1 makes a strictly negative, both at positive noise and at zero, contradicting $a \geq 0$. □

Theorem 7.2 (Conditional global conclusion)**.** *Assume $C_j \leq C_{\mathrm{E}}$ for $1 \leq j \leq 699$, the stated external structural and binomial theorems, and the complete entry predicates at every least failure above this base. Then $C_n < C_{\mathrm{E}}$ for $n \geq 700$ and $\sup_{n \geq 1} C_n = C_{\mathrm{E}}$.*

Proof. A least failure of the strict bound has a true sharp predecessor by the base and minimality. Attainment and the closed-event reflection argument supply its maximizing pair with $a \geq 0$; the preceding theorem excludes it. The independent limit (2) proves sharpness. □

A base only through 41 instead locates any first failure in $42, \ldots, 699$. An unresolved earlier failure prevents the predecessor argument at later indices. Thus no unconditional cutoff at 700 or complete finite-base proof is asserted.

Reproducibility and AI assistance.

The selected entry retains two coarse passes, four signed passes, 11264 nominal boxes and its internal 100-panel incidence contract. The one-strip argument replaces only downstream branches. Exact checks and actual-law controls supplement the proofs; they do not establish the missing finite base. Sources, correspondence and check summaries accompany this manuscript in PR 6 of `https://github.com/kolodziejek/berry-esseen-certificates`. AI systems assisted extensively with research, programs, drafting and component review. The authors remain responsible; AI agreement is not mathematical verification. No full Lean formalization is claimed.

A The uniform Fourier enclosure

Fix an actual index $n \geq N \geq 42$, a centered variance-one marginal X and raw threshold H. Their values may vary with n. Write

$$f(t) = \mathbb{E}e^{itX}, \quad \gamma = \mathbb{E}X^3, \quad e_n(t) = e^{-nt^2/2}(1 - in\gamma t^3/6), \quad V_n = f^n - e_n,$$

$$I_0 = \int_0^{1/2} \frac{|\Im(e^{-itH}V_n)|}{t^3}dt, \qquad I_\infty = \int_{1/2}^{\infty} \frac{|V_n|}{t^3}dt.$$

Compensated inversion.

Lemma A.1. *For a finite signed measure μ of mass and first moment zero with $\int x^2 d|\mu| < \infty$, every $H \in \mathbb{R}$ satisfies*

$$\int (x - H)|x - H|d\mu(x) = -\frac{4}{\pi}\int_0^\infty \frac{\Im(e^{-itH}\widehat{\mu}(t))}{t^3}dt,$$

with absolute convergence.

Proof. Integration by parts gives $\int_0^\infty (ty - \sin ty)t^{-3}dt = \pi y|y|/4$. Indeed positive Tonelli computes its constant from $\int_0^\infty (1 - \cos t)t^{-2}dt = \int_0^\infty s\{s^{-1} - s/(s^2 + 1)\}ds = \pi/2$. The compensated integrand has the sign of y; its absolute double integral for $y = x - H$ is $\pi \int (x - H)^2 d|\mu|/4 < \infty$. Absolute Fubini and $\int (x - H)d\mu = 0$ prove the claim, including atoms. □

Apply this to the sum law minus the scale-$\sqrt{n}$ pushforward of $\phi(x)\{1 + \gamma(x^3 - 3x)/(6\sqrt{n})\}dx$. The comparator has mass one, mean zero and transform e_n. Gaussian integration by parts, with $q_z''' = 4\delta_z$, gives the exact core remainder and its bound:

$$E_n = \frac{2\sqrt{n}}{\pi}\int_0^\infty \frac{\Im(e^{-itH}V_n)}{t^3}dt, \qquad |E_n| \leq \frac{2\sqrt{n}}{\pi}(I_0 + I_\infty). \tag{59}$$

The complete low-frequency bound.

Assume common actual ceilings $\mu_4 \leq M \leq 111/10$, $|\gamma| \leq g \leq 37/20$, $\mu_6 \leq M_6$, $|(3 - \mu_4)/24| \leq A$, $|(10\gamma - \mathbb{E}X^5)/120| \leq Z$. Put

$$E = M_6/720 + g^2/72 + (1/2 + g^2/144)^3/5, \qquad B = A + Z/2 + E/4,$$
$$a_0 = (N - 1)/(2N), \quad b_0 = (45 - M)(1 - 2/N)/96, \quad \Lambda_N = \sqrt{\pi/2}(N/(N - 1))^{3/2}.$$

Proposition A.2. *For $c_0 \geq \Lambda_N$, $c_1 \geq \Lambda_N/2$,*

$$I_0 \leq \frac{L_0 + L_H|H|}{\sqrt{n}} + \frac{L_2}{n},$$

where

$$L_0 = c_0 Z + c_1 Ag + 251g^3/17280, \quad L_H = c_0 A + 251g^2/4800,$$
$$L_2 = E/(2a_0^2) + B^2/(2b_0^3).$$

Proof. For $h = e^{-t^2/2 - i\gamma t^3/6}$ and $0 \leq t \leq 1/2$, Taylor's integral remainder gives

$$f - h = -\frac{3 - \mu_4}{24}t^4 - i\frac{10\gamma - \mathbb{E}X^5}{120}t^5 + r_6, \quad |r_6| \leq Et^6, \quad |f - h| \leq Bt^4. \tag{60}$$

For $w = -t^2/2 - i\gamma t^3/6$, use $|w| \le t^2(1/2 + g^2/144) < 1/6$ and $|e^w - 1 - w - w^2/2| \le |w|^3/5$; the sixth-degree term of $w^2/2$ costs $g^2t^6/72$, and the marginal remainder costs $\mu_6 t^6/720$. Four integrations of $1 - \cos u \ge 0$ give $\cos u \le 1 - u^2/2 + u^4/24$. Symmetrization implies

$$|f|^2 \le 1 - t^2 + (M+3)t^4/12 \le 1 - (45 - M)t^2/48.$$

Both f, h, and their segment, thus have modulus at most $e^{-(45-M)t^2/96}$. Taylor expansion of the polynomial v^n gives

$$f^n - h^n = n(f-h)h^{n-1} + R_2, \quad |R_2| \le \tfrac{1}{2}n(n-1)B^2t^8e^{-(n-2)(45-M)t^2/96}.$$

Integrating r_6 and R_2 after division by t^3 costs L_2/n: $n-1 \ge 2a_0n$, $(n-2)(45-M)/96 \ge b_0n$, and $\int_0^\infty t^3e^{-At^2}dt = 1/(2A^2)$, $\int_0^\infty t^5e^{-At^2}dt = A^{-3}$. For $b = n-1$, the remaining rotated leading imaginary part is bounded by $ne^{-bt^2/2}\{(Z + A|H|)t^5 + bAgt^7/6\}$. Its integral is at most $\Lambda_N(Z + A|H| + Ag/2)/\sqrt{n}$. Finally, putting $u = n\gamma t^3/6$ gives

$$|\Im(e^{-itH}(h^n - e_n))| \le e^{-nt^2/2}(|u|^3/6 + |H|tu^2/2).$$

Gaussian moments yield $\sqrt{\pi/2}\{5g^3/432 + g^2|H|/24\}/\sqrt{n}$; $\sqrt{\pi/2} < 251/200$ proves the stated coefficients. All enlarged integrands are nonnegative. □

The sharper fifth-order input follows from an exact moment identity. For acquired support $|X| \le R \le 6$, $R \ge 1$, and $\mathbb{E}|X|^5 \le E_5$, take $M_6 = RE_5$ and $A = \max(2, M-3)/24$. For $Q = X^2 - \gamma X - 1$,

$$\mathbb{E}Q^2 = \mu_4 - 1 - \gamma^2, \qquad \mathbb{E}X^5 = \gamma^3 + 2\gamma + \mathbb{E}[Q^2(X + 2\gamma)].$$

Reflect this auxiliary moment calculation so $t = |\gamma|$. For $r \ge \max(R, 2g)$ both signs of $\mathbb{E}X^5 - 10\gamma$ are bounded by

$$P_r = r(M-1) + (10 - 2M)t - rt^2 + t^3, \quad Q_r = r(M-1) + (2M-10)t - rt^2 - t^3.$$

The coefficients of μ_4 are $r \mp 2t \ge 0$. If $M \le 5$, $P_r \ge Q_r$ and $P_r \le r(M-1) + (10-2M)t - (r-g)t^2$. If $M > 5$, $P_r \le r(M-1)$ and $Q_r \le r(M-1) + (2M-10)t - rt^2$. The quadratic maxima use vertices clipped to $[0, g]$. For $r = 6$, $P_6'' = -12 + 6t < 0$: when $M \le 5$ and $P_6'(g) \ge 0$ use $P_6(g)$; otherwise use $6(M-1) + (10-2M)^2/[4(6-g)]$. For $M > 5$ also use $6(M-1) + (2M-10)^2/24$. Minimizing the applicable ceilings with $E_5 + 10g$ and dividing by 120 therefore gives the claimed valid Z.

Every high-frequency piece and its floor gate.

For two iid copies assume an actual event $\mathcal{E}$ with $\mathbb{P}(\mathcal{E}) \ge \alpha > 0$, $D = |X - X'| \ge d > 0$ on $\mathcal{E}$, and $\mathbb{E}[D^2 \mid \mathcal{E}] \le M_E$. Fix $b = 9/10$, $0 < r \le 1/4$, and set

$$h = 333/106 - (22/7)^3r^2/6 > 0, \quad \kappa = \alpha(23/48)^2d^2, \quad L(x) = (x+2)/(x^2+4x+2),$$
$$S_3 = \sum_{j=1}^{20} j^{-3} + 1/800, \quad S_5 = \sum_{j=1}^{20}(j-r)^{-5} + [4(20-r)^4]^{-1},$$
$$c(u) = u/2 - (M+3)u^2/24, \quad c_* = \min\{c(1/4), c(81/100)\},$$
$$c_q = 1/8 - (M+3)/384, \qquad s = [1 - (M+3)53/600]/2.$$

Proposition A.3. *If $c_* > 0$, $\sqrt{n} I_\infty \le \sum_{i=1}^{6} T_i(n)$, where*

$$T_1 = 2\sqrt{n} e^{-nc_*}, \qquad T_2 = \frac{M_E}{4\pi^{3/2} h\sqrt{\alpha}} \{S_3 + 3S_5/(n\alpha h^2)\},$$

$$T_3 = \frac{M_E \sqrt{n}}{2\pi^2} e^{-n\alpha h^2 r^2}, \qquad T_4 = \frac{e^{-n\kappa b^2}}{2b^4 \kappa \sqrt{n}},$$

$$T_5 = \frac{\sqrt{n}}{2b^2} e^{-n\alpha (23/48)^2}, \qquad T_6 = \sqrt{n} e^{-n/8} \{2L(n/8) + \sqrt{M-1}/3\}.$$

The optional replacements, minimized with the respective simple terms, are

$$\widetilde{T}_1 = 2\sqrt{n} e^{-nc_q} L(ns/4) \quad (s > 0), \qquad \widetilde{T}_4 = \frac{\sqrt{n} e^{-n\kappa b^2} L(n\kappa b^2)}{2b^2}.$$

The common-floor sum at N is valid if

$$2Nc' > 1 \quad (c' \in \{c_*, \alpha h^2 r^2, \alpha (23/48)^2, 1/8\}).$$

The optional first term requires $2Nc_q > 1$, and the optional fourth $N\kappa b^2 \ge 1/4$. Without an optional gate, retain the simple term.

Proof. Separate $|f|^n$ and the entire comparator. On $[1/2, b]$, $|f|^n \le e^{-nc(t^2)} \le e^{-nc_*}$ and $\int t^{-3} dt < 2$, giving T_1. Its optional chord follows from $c(u) - c_q - s(u - 1/4) = (M+3)(u - 1/4)(81/100 - u)/24 \ge 0$. For $x = nab^2 > 0$ the integration formula is

$$\sqrt{n} \int_b^\infty e^{-nat^2} t^{-3} dt \le \frac{\sqrt{n} e^{-x}}{2b^2} L(x). \tag{61}$$

Substitution and positive Fubini give the left side as this prefactor times $1 - xQ(x)$, where $Q(x) = \int_0^\infty e^{-u}/(x+u) du$. For $P(u) = u^2 - 4u + 2$ and $D_x = P(-x) = x^2 + 4x + 2$, use $P(u) - D_x = (u+x)(u-x-4)$ and $\int_0^\infty P(u) e^{-u} du = \int_0^\infty u P(u) e^{-u} du = 0$. Two divisions therefore give

$$Q(x) - \frac{x+3}{D_x} = \frac{1}{D_x^2} \int_0^\infty \frac{P(u)^2 e^{-u}}{x+u} du \ge 0.$$

Thus $1 - xQ(x) \le L(x)$, proving the formula and the optional chord term.

For $t \ge b$, symmetrization and convex Jensen give

$$|f(t)|^n \le e^{-n\mathbb{E} \sin^2(t(X-X')/2)} \le \mathbb{E}[e^{-n\alpha \sin^2(tD/2)} \mid \mathcal{E}].$$

Positive Tonelli permits a separate partition for each D: with $\ell = 2\pi/D$, use $[0, \ell/2]$ and nonzero cells $t = \ell(j+v)$, $j \ge 1$, $|v| \le 1/2$. On $|v| \le r$, $|\sin \pi v| \ge h|v|$; on the wings it is at least hr, by $\sin x \ge x - x^3/6$ and monotonicity on $[0, \pi/2]$. Taylor's remainder gives

$$\{(j-v)^{-3} + (j+v)^{-3}\}/2 \le j^{-3} + 6(j-r)^{-5} v^2.$$

Extend the Gaussian core integral to $\mathbb{R}$. Its zeroth and second moments, the Jacobian $t^{-3} dt = D^2 (j+v)^{-3} dv/(4\pi^2)$ and the conditional second-moment bound give T_2; decreasing-integral tails beyond 20 give S_3, S_5. The wing weight is at most $\int_{\ell/2}^\infty t^{-3} dt = D^2/(2\pi^2)$, giving T_3. Extending these sets below b only increases their bounds. In the zero cell, $u = tD/2 \le \pi/2$. On $u \le 1/2$, $\sin u \ge 23u/24$ and $D \ge d$. Integrate over the enlarged $[b, \infty)$ with $t^{-3} \le t/b^4$ to obtain T_4; (61) gives its optional form. On $u \ge 1/2$, $\sin u \ge 23/48$ gives T_5. Cell endpoints have zero Lebesgue measure even when D has atoms. For the comparator,

$|e_n| \le e^{-nt^2/2}(1 + n|\gamma|t^3/6)$. Use (61) at $b = 1/2$ and $\int_{1/2}^{\infty} e^{-nt^2/2}dt \le 2e^{-n/8}/n$; $\gamma^2 \le M - 1$ from $\mathbb{E}(X^2 - \gamma X - 1)^2 \ge 0$ gives T_6. No comparator or lattice tail is omitted.

For floor evaluation, $\sqrt{n}e^{-nc'}$ decreases when $2nc' > 1$, whereas n^{-1} and $n^{-1/2}e^{-nc'}$ decrease for $c' > 0$. Also $L'(x) = -(x^2 + 4x + 6)/(x^2 + 4x + 2)^2 < 0$. The derivative of $\sqrt{x}e^{-x}L(x)$ has the sign of $4 - 10x - 22x^2 - 13x^3 - 2x^4$, negative and decreasing for $x \ge 1/4$. These prove every stated gate on a common scalar majorant, not by comparing characteristic functions of different laws. □

For any complete HIGH sum K_H at N, the final bound is

$$|E_n| \le \frac{2}{\pi}\{L_0 + K_H + L_2/\sqrt{N} + L_H|H|\} \le \frac{16}{25}\{L_0 + K_H + L_2/\sqrt{N} + L_H|H|\}. \tag{62}$$

Complete HIGH windows may be minimized; the LOW triple stays complete. Round denominator roots down and numerator roots up, using the constants below. For $x \ge 0$, choose rational $0 \le y \le \min(x, 64)$ and round $1/\sum_{j=0}^{64} y^j/j!$ upward to majorize e^{-x}. The floor proof concerns the analytical functions, not monotonicity of this rounded exponential helper.

B Elementary constants

Lemma B.1. *The standard normal density satisfies*

$$\|\phi''\|_\infty = \phi(0) < 2/5, \quad \|\phi'''\|_\infty \le 2/\pi, \quad 1/(4\pi) < 2/25,$$

$$819/2000 < C_{\mathrm{E}} < 20487/50000 < 41/100, \qquad \phi(0) > .39886 > .3988 = 997/2500.$$

Moreover $333/106 < \pi < 22/7$, $\sqrt{\pi} > 4431/2500$, *and* $0.4097321837023 < C_{\mathrm{E}} < 0.4097321837025$.

Proof. For $J(t) = \int e^{-x^2/2}e^{itx}dx$, integration by parts gives $J' = -tJ$; the squared Gaussian integral in polar coordinates gives $J(0) = \sqrt{2\pi}$. Thus $\phi(x) = (2\pi)^{-1}\int e^{-t^2/2}e^{-itx}dt$. Differentiating under integrable majorants gives $\|\phi'''\|_\infty \le (2\pi)^{-1}\int |t|^3 e^{-t^2/2}dt = 2/\pi$; likewise $\|\phi^{(4)}\|_\infty \le 3\phi(0)$ and $\|\phi^{(5)}\|_\infty \le 8/\pi$. The identity $\phi'' = (x^2 - 1)\phi$ gives its exact norm: the exterior maximum is $2e^{-3/2}\phi(0) < \phi(0)$.

Put $s_k(x) = \sum_{j=0}^{k-1}(-1)^j x^{2j+1}/(2j + 1)$. Tangent addition and the bounds $0 < 4(1/5 - 1/(3 \cdot 5^3)) - 1/239 < 4\arctan(1/5) - \arctan(1/239) < 1 < \pi/2$ fix the first-quadrant branch (use $\pi/2 = 2\int_0^1 (1 + t^2)^{-1}dt > 1$). Hence Machin's identity is $\pi = 16\arctan(1/5) - 4\arctan(1/239)$. Integrating the finite geometric expansion of $(1 + t^2)^{-1}$ gives

$$\pi_- := 16s_{12}(1/5) - 4s_3(1/239) < \pi < 16s_{11}(1/5) - 4s_4(1/239) =: \pi_+.$$

Exact rational comparisons give $333/106 < \pi_- < \pi_+ < 22/7$ and $\pi_- > (4431/2500)^2$. Use the rational endpoints

$$r_- = 3162277660168379/10^{15}, \qquad r_+ = 3162277660168380/10^{15},$$

$$c_- = .4097321837023, \qquad c_+ = .4097321837025.$$

Squaring gives $r_-^2 < 10 < r_+^2$; direct integer multiplication gives

$$c_-^2 < \frac{19 + 6r_-}{72\pi_+} < C_{\mathrm{E}}^2 < \frac{19 + 6r_+}{72\pi_-} < c_+^2.$$

This licenses the fine entry endpoints; the displayed coarse bounds for C_{E} follow since $819/2000 < c_- < c_+ < 20487/50000 < 41/100$. All denominators are positive and explicit. Finally $\pi > 25/8$ gives the coarse upper bounds, while $\pi < 22/7$ and $(.39886)^2 < 7/44$ give the lower bound for $\phi(0)$. □

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